COVER SHEET FOR PROPOSAL TO THE NATIONAL SCIENCE FOUNDATION

PROGRAM ANNOUNCEMENT/SOLICITATION NO./ DUE DATE
NSF 26-520

☐ Special Exception to Deadline Date Policy

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2701227

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MCP - ALGEBRA,NUMBER THEORY,AND COM

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NAME OF PRIMARY PLACE OF PERF
Dartmouth College

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TITLE OF PROPOSED PROJECT
Multigraded Homological Algebra and Geometry

SHOW LETTER OF INTENT ID IF APPLICABLE

REQUESTED AMOUNT	PROPOSED DURATION (1-60 MONTHS)	REQUESTED STARTING DATE	SHOW RELATED PRELIMINARY PROPOSAL NO. IF APPLICABLE
$ **373,726**	24 months	**05/01/2027**	

THIS PROPOSAL INCLUDES ANY OF THE ITEMS LISTED BELOW

☒ TYPE OF PROPOSAL **Research**
☒ COLLABORATIVE STATUS **Non-Collaborative**
☐ BEGINNING INVESTIGATOR
☐ DISCLOSURE OF LOBBYING ACTIVITIES
☐ PROPRIETARY & PRIVILEGED INFORMATION
☐ HISTORIC PLACES
☐ LIVE VERTEBRATE ANIMALS IACUC App. Date
PHS Animal Welfare Assurance Number

☐ HUMAN SUBJECTS Human Subjects Assurance Number
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☐ POTENTIAL LIFE SCIENCES DUAL USE RESEARCH OF CONCERN
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CERTIFICATION PAGE

Certification for Authorized Organizational Representative (or Equivalent)
By electronically signing and submitting this proposal, the Authorized Organizational Representative (AOR) is: (1) certifying that statements made herein are true and complete to the best of the individual's knowledge; and (2) agreeing to accept the obligation to comply with NSF award terms and conditions if an award is made as a result of this proposal. Further, the proposer is hereby providing certifications regarding conflict of interest, flood hazard insurance, responsible and ethical conduct of research, organizational support, and safe and inclusive working environments for off-campus or off-site research, as set forth in the NSF Proposal & Award Policies & Procedures Guide (PAPPG). Willful provision of false information in this application and its supporting documents or in reports required under an ensuing award is a criminal offense (U.S. Code, Title 18, Section § 1001).

Certification Regarding Conflict of Interest
The AOR is required to complete certifications stating that the organization has implemented and is enforcing a written policy on conflicts of interest (COI), consistent with the provisions of PAPPG Chapter IX.A.; that, to the best of the individual's knowledge, all financial disclosures required by the conflict of interest policy were made; and that conflicts of interest, if any, were, or prior to the organization's expenditure of any funds under the award, will be, satisfactorily managed, reduced or eliminated in accordance with the organization's conflict of interest policy. Conflicts that cannot be satisfactorily managed, reduced or eliminated and research that proceeds without the imposition of conditions or restrictions when a conflict of interest exists, must be disclosed to NSF via use of the Notifications and Requests Module in Research.gov.

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(Applies to Institutions of Higher Education only and is not applicable to proposals for conferences, symposia, and workshops**)**
By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that the institution has a plan in place to provide appropriate training and oversight in the responsible and ethical conduct of research to undergraduate students, graduate students, postdoctoral researchers, faculty, and other senior/key personnel who will be supported by NSF to conduct research and that such training addresses mentor training and mentorship, training to raise awareness of potential research security threats, and Federal export control, disclosure, and reporting requirements.

The AOR shall require that the language of this certification be included in any award documents for all subawards at all tiers.

Certification Regarding Organizational Support
By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is certifying that there is organizational support for the proposal as required by Section 526 of the America COMPETES Reauthorization Act of 2010. This support extends to the portion of the proposal developed to satisfy the Broader Impacts Review Criterion as well as the Intellectual Merit Review Criterion, and any additional review criteria specified in the solicitation. Organizational support will be made available, as described in the proposal, in order to address the broader impacts and intellectual merit activities to be undertaken.

Certification Regarding Dual Use Research of Concern
By electronically signing the certification pages, the Authorized Organizational Representative is certifying that the organization will be or is in compliance with all aspects of the United States Government Policy for Institutional Oversight of Life Sciences Dual Use Research of Concern.

Certification Requirement Specified in the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, Section 223(a)(1) (42 USC 6605(a)(1))
By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that each individual employed by the organization and identified on the proposal as senior/key personnel has been made aware of the certification requirements identified in the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, Section 223(a)(1) (42 USC 6605(a)(1)).

Certification Regarding Safe and Harassment-free Working Environments for Off-Campus or Off-Site Research
(This certification applies only to proposals in which data/information/samples are being collected off-campus or off-site, such as fieldwork and research activities on vessels and aircraft.)
By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in accordance with the NSF Proposal & Award Policies and Procedures Guide, Chapter II.E.9, the organization has a plan in place for this proposal regarding safe and harassment-free working environments.

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By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. 19232), all individuals identified as senior/key personnel have been made aware of and have complied with their responsibility under that section to certify that the individual is not a party to a malign foreign talent recruitment program. Additionally, individuals serving as a Principal Investigator (PI) or co-PI on an active NSF award made on or after May 20, 2024, are aware they must certify annually via Research.gov to their participation or non-participation in an MFTRP. Additional information regarding the annual certification process can be found at nsf.gov/research-security.

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By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), the AOR is certifying that all individuals identified as senior/key personnel have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certification Regarding IHEs Hosting or Supporting Confucius Institutes
By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in compliance with Section 10339A of the CHIPS and Science Act of 2022 (42 U.S.C. § 19039), the AOR is certifying that, absent a waiver granted by the NSF Director, the IHE does not maintain a contract or agreement between the institution and a Confucius Institute.

AUTHORIZED ORGANIZATIONAL REPRESENTATIVE	SIGNATURE	DATE
NAME **Arlanda Erzen Lumbra**	**Electronic Signature**	**Oct 08 2026 04:01 PM**

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Project Summary: Multigraded Homological Algebra & Geometry

Overview.
Syzygies, the relations among equations, encode algebra and geometry, but much of the classical theory applies works best for standard graded rings. Rings with weighted or multiple gradings are unavoidable in modern commutative algebra and geometry: modular curves, section rings of stacky curves, and Cox rings of toric varieties all require multi/nonstandard gradings. This project develops multigraded homological algebra in three directions: (i) the syzygies of stacky curves, aimed at an analog of Green's conjecture for stacky curves; (ii) the shape and bounds of multigraded Castelnuovo–Mumford regularity and its connections to Betti numbers and toric geometry; and (iii) a regularity theory on the Cox category that repairs fragilities of multigraded regularity. New *Macaulay2* tools, mentoring, educational resources, and organizing complement the research.

Intellectual Merit.
The first direction develops syzygies of stacky curves, where fractional divisors play the role classically played by degree. The PI will determine the generator degrees of stacky canonical rings, sharpening Voight and Zureick-Brown's stacky Noether–Petri theorem, and relate their syzygies to those of the coarse curve. Since stacky points enlarge the Betti table beyond its classical shape, the stacky analog of Green's conjecture splits in two: a weak form, predicting that the coarse curve controls the linear strand and weighted N_p-properties, and a strong form, predicting that the full Betti table depends only on the coarse curve and the signature. Further projects study high-degree line bundles and log canonical rings on stacky curves.

The second direction explains the shape of multigraded regularity on smooth projective toric varieties, where regularity is a region the Betti table does not determine. The PI proposes that support geometry governs the directions in which this region opens and that syzygies govern where its boundary lies. Together, these ideas will clarify how regularity relates to Betti numbers and yield the first finite algorithm for computing a module's full regularity region. A central goal is a toric analog of the Gruson–Lazarsfeld–Peskine theorem, which bounds the regularity of a projective curve by its degree: the PI conjectures an explicit bound in terms of the curve's numerical class and combinatorial data of the ambient variety. Finally, the PI will seek combinatorial formulas for the regularity of torus-invariant subvarieties, which extend to bounds on arbitrary subvarieties.

The third direction asks whether multigraded regularity is the right definition in the wrong category. Working in the Cox category, which unifies all toric birational models sharing a Cox ring, the PI uses the uniform Hanlon–Hicks–Lazarev (HHL) resolution and the Bondal–Thomsen collection to propose a new definition of regularity that recovers the classical one on projective space and makes the structure sheaf $\mathbf{0}$-regular on every smooth projective model. Central questions in this project are whether regularity in the PI's sense persists under nef twists, whether it implies global generation, and how it relates to Maclagan–Smith regularity.

Two software projects yield community tools: further development of the *Macaulay2* package `toricStacks` and a database of the expert-chosen examples in *Macaulay2*'s documentation, tracking authorship, as a first step toward community-centered AI/ML tools.

Broader Impacts.
The PI advises three Ph.D. students on problems central to the proposal and will continue mentoring undergraduates. The PI will turn her Preliminary Arizona Winter School lectures and commutative algebra course notes into a modern, inquiry-based introductory book on algebraic geometry and commutative algebra. The PI is leading a `toricStacks` group at the 2027 *Macaulay2* conference and will host *Gems of Commutative Algebra III* in Fall 2028. The PI has submitted a proposal to organize a 2030 Fields thematic semester in combinatorial algebraic geometry. The PI will continue organizing *Unwritten Rules* and working with Dartmouth's AWM chapter. The PI's work on AMS committees and the *Spectra* Board of Trustees extends these efforts nationally.

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Project Description: Multigraded Homological Algebra and Geometry

Overview. Castelnuovo–Mumford regularity and graded Betti numbers link the geometry of a projective variety to the algebra of its equations. This proposal carries this theory beyond the standard grading: to nonstandard $\mathbb{Z}$-gradings, which arise from stacky curves, and to $\mathbb{Z}^r$-gradings, which arise from toric varieties. After reviewing results of prior NSF support (§1), we develop a Green's-conjecture-type theory for stacky canonical curves (§2); structural results and geometric bounds for multigraded regularity (§3); a uniform regularity theory on the Cox category of a toric variety (§4); and open-source software (§5), with broader impacts in (§6).

1. Results from Prior NSF Support

1.1 Intellectual Merit from Prior NSF Support From 2020 to 2022, the PI held an NSF Postdoctoral Research Fellowship (MSPRF DMS-2002239, $150,000), *Asymptotic Syzygies in Algebraic Geometry.* As publication lags the award, these figures span 2020–2026: 10 arXiv preprints [39, 41, 40, 28, 27, 7, 42, 8, 38, 37] and 8 published articles, including in *Geometry & Topology* [27], *Advances in Mathematics* [41], *Journal of Algebra* [39], and *Experimental Mathematics* [43]. This work advanced multigraded homological algebra (§1.2) and the cohomology of moduli spaces (§1.4). Its data and software, e.g., *Syzygy Data* [43, 39], are publicly available.

1.2 Eisenbud–Goto on Products of Projective Spaces The Castelnuovo–Mumford regularity of a projective variety $X \subset \mathbb{P}^r$ measures the complexity of X through the vanishing of its cohomology. Its centrality rests on the fact that a single integer controls cohomology vanishing, the degrees of generators and syzygies, and the eventual linearity of resolutions.

Theorem 1.1. [69] *Let $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}^n$ and $M = \bigoplus_{e\in\mathbb{Z}} H^0(\mathbb{P}^n, \mathcal{F}(e))$ the corresponding section module. The following are equivalent:*

(1) M is d-regular in the sense of [118, Lect. 14]*;*
(2) $\beta_{p,q}(M) = 0$ for all $p \geq 0$ and $q > d + p$; and
(3) $M_{\geq d}$ has a d-linear resolution in the sense of [69].

Nearly every use of regularity in the literature – regularity of powers of ideals [60, 103], the Eisenbud–Goto conjecture [115], the construction and boundedness of Hilbert schemes [88, 84], Gröbner basis complexity [13] – passes through these equivalences.

A toric variety $X = X_\Sigma$ associated to a fan Σ with rays $\Sigma(1)$ is a well-behaved compactification of a torus and, under mild hypotheses, a quotient $X = (\mathbb{C}^{\Sigma(1)} \setminus \mathbb{V}(B))/(\mathbb{C}^\times)^r$, where B is the *irrelevant ideal* of X; for $X = \mathbb{P}^n$, this is $B = \langle x_0, \ldots, x_n\rangle$. Cox's theorem [57] identifies coherent sheaves on X with B-saturated $\mathbb{Z}^r$-graded modules over the *Cox ring* $S = \mathbb{C}[x_\rho \mid \rho \in \Sigma(1)]$, translating the geometry of X into multigraded commutative algebra. This dictionary is far subtler than for $\mathbb{P}^n$, driving much current work: virtual resolutions and resolutions of the diagonal [18, 94, 33, 77], generalizations of the BGG correspondence [68, 35], and multigraded local cohomology [25, 51].

Extending regularity beyond projective space has a long history: Benson's theory for nonstandard $\mathbb{Z}$-graded rings [16], developed by Symonds in his groundbreaking work on invariant theory [129, 130], and Hoffman–Wang's for products of two projective spaces [97]. Maclagan and Smith's cohomological definition [111] subsumes both and is the standard framework on toric varieties. Throughout, X denotes a smooth projective toric variety, so that $\mathrm{Pic}(X) \cong \mathbb{Z}^r$ and the nef cone $\mathrm{Nef}(X)_\mathbb{R} \subset \mathrm{Pic}(X)_\mathbb{R}$ is full-dimensional. We write $\mathrm{Nef}(X) \coloneqq \mathrm{Nef}(X)_\mathbb{R} \cap \mathrm{Pic}(X)$ for the nef monoid and $B_1, \ldots, B_s$ for its unique minimal generators. For an integer vector $\mathbf{v} = (v_1, \ldots, v_t)$ of any length, set $|\mathbf{v}| \coloneqq v_1 + \cdots + v_t$.

Definition 1.2. [111, Def. 6.6] *A coherent sheaf $\mathcal{F}$ on X is* $\mathbf{d}$-regular*, for $\mathbf{d} \in \mathbb{Z}^r$, if, for all $i > 0$,*

$$H^i(X, \mathcal{F}(\mathbf{e})) = 0 \qquad \text{for all} \qquad \mathbf{e} \in L_i(\mathbf{d}) \coloneqq \bigcup_{\mathbf{v}\in\mathbb{N}^s, |\mathbf{v}|=i} \left(\mathbf{d} - \sum_{j=1}^{s} v_j B_j + \mathrm{Nef}(X) \right).$$

The multigraded regularity *of* $\mathcal{F}$ *is* $\operatorname{reg}(\mathcal{F}) \coloneqq \{\mathbf{d} \in \mathbb{Z}^r \mid \mathcal{F} \text{ is } \mathbf{d}\text{-regular}\}$.

A graded module M is $\mathbf{d}$-regular if $H^i_B(M)_{\mathbf{e}} = 0$ for $i > 0$, $\mathbf{e} \in L_{i-1}(\mathbf{d})$, and $H^0_B(M)_{\mathbf{e}} = 0$ for $\mathbf{e} \in \bigcup_j (\mathbf{d} + B_j + \operatorname{Nef}(X))$, where $L_0(\mathbf{d}) = \mathbf{d} + \operatorname{Nef}(X)$. Multigraded regularity is thus not a single number but an infinite subset of $\mathbb{Z}^r$, whose boundary can be intricate (Figure 1).

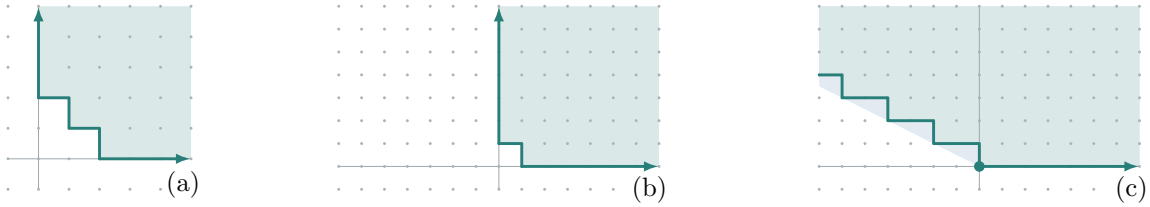


FIGURE 1. Regularity regions of (a) S/I_Z for three points $Z = \{([1:1],[1:4]),([1:2],[1:5]),([1:3],[1:6])\} \subset \mathbb{P}^1 \times \mathbb{P}^1$; (b) $\mathcal{O}_{\mathcal{H}_2}$ here $\mathcal{H}_2$ is the second Hirzebruch surface; (c) S/I_p for a point $p \in \mathcal{H}_2$.

Multigraded regularity has become a central tool in the study of uniform bounds and Hilbert schemes [112, 5], syzygies and local cohomology [126, 95, 25], virtual resolutions [18, 41], and asymptotic regularity [91, 42]. Yet for more than fifteen years after its introduction, a foundational question remained open: *Is there a multigraded Eisenbud–Goto theorem?* That is, can multigraded regularity be characterized by free resolutions and truncations, as in Theorem 1.1? Direct generalizations fail even for products of projective spaces. Answering a question of Maclagan–Smith, further studied by Botbol–Chardin [25], the PI and coauthors showed that multigraded Betti numbers do not determine regularity [41, Ex. 5.1], ruling out an analogue of condition (2). Likewise, the truncation $M_{\geq \mathbf{d}}$ of a $\mathbf{d}$-regular module M need not have a linear resolution, so condition (3) must also be reformulated.

A major advance came with Berkesch, Erman, and Smith's characterization of multigraded regularity on products of projective spaces via virtual resolutions with controlled degrees [18, Thm. 2.9], a counterpart to condition (2). Building on these ideas, the PI and coauthors introduced *quasilinear resolutions* and proved a multigraded Eisenbud–Goto theorem in which quasilinear resolutions of truncations play the role of condition (3).

Theorem 1.3. [41, Thm. A] *Let* $\mathcal{F}$ *be a coherent sheaf on* $X = \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_r}$ *and* $M = \bigoplus_{\mathbf{e} \in \mathbb{Z}^r} H^0(X, \mathcal{F}(\mathbf{e}))$ *the corresponding section module. For* $\mathbf{d} \in \mathbb{Z}^r$*, the following are equivalent:*

(1) M *is* $\mathbf{d}$*-regular in the sense of* [111]*;*
(2) M *has a virtual resolution with degrees as in* [18, Thm. 2.9]*; and*
(3) $M_{\geq \mathbf{d}}$ *has a* $\mathbf{d}$*-quasilinear resolution in the sense of* [41].

When $r = 1$, Theorem 1.3 recovers the classical Eisenbud–Goto theorem, and (1)⇔(3) holds for every finitely generated M with $H^0_B(M) = 0$. Equally important are its methods: rather than resolving $M_{\geq \mathbf{d}}$ directly, we compute its multigraded Betti numbers via a spectral sequence arising from the Fourier–Mukai transform of $\widetilde{M}$ with Beilinson's resolution of the diagonal [15] as kernel. This "resolution of the diagonal" approach, now a central technique [35, 33, 94, 36], also yields an effective algorithm [3, 59] for testing $\mathbf{d}$-regularity and producing short virtual resolutions.

1.3 Bounds & Asymptotic Behavior of Multigraded Regularity of Ideals Partially order $\operatorname{Pic}(X)$ by $\mathbf{d} \leq \mathbf{e}$ when $\mathbf{e} - \mathbf{d} \in \operatorname{Nef}(X)$. Although nef twists preserve regularity, $\operatorname{reg}(M)$ need not have finitely many minimal elements (pictorially, corners). In [40, Ex. 1.1], the PI, Cranton Heller, and Sayrafi exhibit a cyclic module on a Hirzebruch surface with infinitely many minimal degrees. We show that for faithful modules such pathologies cannot occur.

Theorem 1.4. [40, Cor. 4.6] *Let* M *be a finitely generated graded* S*-module with* $\operatorname{ann}(M) = 0$ *and* $\widetilde{M} \neq 0$*. If its homogeneous generators* f_i *satisfy* $\mathbf{q} \leq \deg f_i$*, then* $\operatorname{reg}(M) \subseteq \mathbf{q} + \operatorname{Nef}(X)$*. Consequently,* $\operatorname{reg}(M)$ *has finitely many minimal elements.*

The proof uses the fact that $\mathbf{d}$-regularity forces $M_{\geq \mathbf{d}}$ to be generated in degree $\mathbf{d}$; a chamber-and-determinant argument turns a violation of the bound into a nonzero annihilator of M. It applies to every nonzero homogeneous ideal $I \subseteq S$, though $\operatorname{reg}(S/I)$ may have infinitely many corners.

Classically, the regularity of powers is linearly bounded (Swanson [128]) and eventually linear (Cutkosky–Herzog–Trung, Kodiyalam [60, 103]). Combining our bound with a resolution of the Rees algebra gives the first multigraded analogue: regularity of powers translates at a linear rate.

Corollary 1.5. [40, Thm. 5.1] *Let $I = \langle f_1, \ldots, f_k \rangle \subseteq S$ be a nonzero homogeneous ideal. There is $\mathbf{a} \in \operatorname{Pic}(X)$ such that for all integers $t > 0$ and all $\mathbf{q}_1, \mathbf{q}_2 \in \operatorname{Pic}(X)$ with $\mathbf{q}_1 \geq \deg f_i \geq \mathbf{q}_2$ for all i,*

$$t\mathbf{q}_1 + \mathbf{a} + \operatorname{reg}(S) \subseteq \operatorname{reg}(I^t) \subseteq t\mathbf{q}_2 + \operatorname{Nef}(X).$$

1.4 Top-Weight Cohomology of $\mathcal{A}_g$ The moduli stack of principally polarized abelian varieties of dimension g, $\mathcal{A}_g \coloneqq [\mathbb{H}_g/\mathrm{Sp}_{2g}(\mathbb{Z})]$, is one of the most classical objects in algebraic geometry, yet its rational cohomology is known completely only for $g \leq 3$ [98, 92], and until recently little was known beyond Borel's stable range [22]. In particular, Grushevsky asked whether $H^{2k+1}(\mathcal{A}_g; \mathbb{Q}) \neq 0$ for some g and k [89]. Chan, Galatius, and Payne's computation of the top-weight cohomology of $\mathcal{M}_g$, combining Deligne's weight filtration with the combinatorics of tropical moduli spaces, opened a powerful new approach to the cohomology of moduli spaces [48] that is now highly active [49, 121, 20, 45]. The PI and collaborators extended this program to $\mathcal{A}_g$. This requires adapting [48] to toroidal compactifications whose boundary strata are indexed by cones in Voronoi's reduction theory. Analyzing these strata and relating the cohomology of $\mathcal{A}_g$ inductively to that of $\mathcal{A}_{g-1}$ and $\mathrm{GL}_g(\mathbb{Z})$ yields many new classes.

Theorem 1.6. [27, Thm. A] *The rational cohomology satisfies $H^k(\mathcal{A}_g; \mathbb{Q}) \neq 0$ for $(g,k) = (5,15)$, $(5,20)$, $(6,30)$, $(7,28)$, $(7,33)$, $(7,37)$, and $(7,42)$.*

Theorem 1.6 answers Grushevsky's question affirmatively and, since $H^*(\mathcal{A}_g; \mathbb{Q}) \cong H^*(\mathrm{Sp}_{2g}(\mathbb{Z}); \mathbb{Q})$, produces new unstable cohomology for the symplectic group. These results inspired a Hopf algebra structure on the top-weight cohomology of $\mathcal{A}_g$ [30] and Brown's bordification of $\mathcal{A}_g^{\mathrm{Trop}}$ [29], both producing further unstable cohomology of $\mathrm{GL}_g(\mathbb{Z})$. The framework of [27] is the template for the PI's subsequent work on tropicalizing general Shimura varieties [7, 8], the subject of a plenary lecture at the 2025 AMS Summer Research Institute in Algebraic Geometry.

1.5 Broader Impacts from Prior NSF Support

1.5.1 Organizing. During and immediately following the award, I organized five conferences: *WAGS* (100 participants), *Gems of Combinatorics I* (40), Spec($\overline{\mathbb{Q}}$) (50), *BATMOBILE 2022* (20), and *Gems of Commutative Algebra II* (40), plus AMS special sessions. The *Gems of* series builds early-career research networks.

1.5.2 Mentoring & Virtual Mathematics. I led four undergraduate algebra and geometry reading courses and volunteered with the *Berkeley Math Circle*, promoting mathematics to high schoolers. In summer 2021, I led six undergraduates in research related to [27] and advised a summer 2022 project on tropical abelian varieties; that undergraduate is now a mathematics Ph.D. student with an NSF GRF. To reach those at the periphery as mathematics moved online during COVID-19, I helped run Ravi Vakil's massive open-access online course *Algebraic Geometry in the Time of Covid* in summer and fall 2020 (∼2,000 participants worldwide) and organized an 8-week virtual undergraduate reading course in commutative algebra in spring 2021.

1.5.3 Community. During 2020–2023, I served on the board of *Spectra*, including as its inaugural president. I oversaw its growth and formalization, including the adoption of bylaws, the creation of an invited lecture at the JMM, and a fundraising campaign that has raised over $20,000.

2. The Geometry of Syzygies on Stacky Curves

Let X be a projective variety and L a very ample line bundle. For the embedding $X \hookrightarrow \mathbb{P}H^0(X,L) \cong \mathbb{P}^n$ defined by L, let $S_X = S/I_X$, where $S = \mathbb{C}[x_0,\dots,x_n]$ is the standard graded polynomial ring and I_X is the homogeneous defining ideal of X. The minimal graded free resolution $0 \leftarrow S_X \xleftarrow{\epsilon} F_0 \xleftarrow{d_1} F_1 \xleftarrow{d_2} \cdots$ of S_X has terms $F_p = \bigoplus_q S(-q)^{\beta_{p,q}(X;L)}$, where $S(-q)_d = S_{d-q}$. The *graded Betti numbers (or syzygies)* $\beta_{p,q}(X;L)$ depend on the extrinsic embedding, not just on X; a central theme of the field is how they nonetheless capture the intrinsic geometry of X.

Forty years after Green's work, this connection is best understood for smooth non-hyperelliptic canonical curves of genus $g \geq 3$ ($L = K_X$). Green's conjecture [87] predicts the vanishing threshold of the quadratic strand:

$$\beta_{p,p+2}(X;K_X) = 0 \quad \Longleftrightarrow \quad p < \mathrm{Cliff}(X), \qquad 0 \leq p \leq g-3. \tag{1}$$

Here $\mathrm{Cliff}(X) \coloneqq \min\{\deg(L) - 2h^0(X,L) + 2 \,:\, h^0(X,L), h^1(X,L) \geq 2\}$, with the convention $\mathrm{Cliff}(X) = 1$ for a non-hyperelliptic curve of genus 3. Voisin's breakthrough proved it for general curves of every genus [134, 135]; new proofs [4, 124, 100, 102] and refinements [76, 52, 74, 75, 66, 101] followed.

This project develops the corresponding theory for *stacky curves* [2, 44, 1], where S is nonstandard graded and its geometry is visible only through fractional divisors. The classical theory of the geometry of syzygies is a theory of standard graded rings, and it does not directly cover section rings requiring generators of different degrees. Already for a point O on an elliptic curve E, the section ring $\bigoplus_n H^0(E,nO)$ needs generators in degrees $1,2,3$ and is cut out by the Weierstrass equation $y^2 = x^3 + axz^4 + bz^6$, a sextic in $\mathbb{P}(1,2,3)$. This is not an edge case but the general picture. By Demazure–Pinkham [122, 65, 136], every finitely generated normal domain R of dimension two with $R_0 = \mathbb{C}$ and an effective positive grading is $\bigoplus_{n\geq 0} H^0(C, \lfloor nD \rfloor)$ for an ample $\mathbb{Q}$-divisor D on a curve C: a section ring on a stacky curve whose stacky points record the fractional part of D. Modular curves are flagship examples; rigidifying $X(1) \cong \mathcal{P}(4,6)$ gives $\mathcal{P}(2,3)$, whose log canonical ring (with the cusp) is $\mathbb{C}[E_4,E_6]$ in degrees $2,3$. Even-weight forms on $X_0(N)$ and Shimura curves give (log) canonical rings, with weight $2d$ in degree d. Voight and Zureick-Brown [133] proved a stacky Noether–Petri theorem presenting these rings (see also [120, 106]), but their syzygies are uncharted: stacky analogues of Green's N_p theorem and Green's conjecture are wide open. This project builds a theory in which fractional divisors play the role that degree plays classically, extending Mumford's and Green's theorems to stacky curves and answering [34, Ques. 7.7]. Results without citations in this section are proved by the PI, M. Banks, and P. Wei.

2.1 Weighted Canonical Rings A *stacky curve* $\mathcal{X}$ is a smooth proper geometrically connected Deligne–Mumford stack of dimension one with a dense open subscheme. (Throughout this section, $\mathbb{K} = \mathbb{C}$.) Concretely, $\mathcal{X}$ is a curve C with finitely many *stacky points* carrying cyclic stabilizers [133, Lem. 5.3.10], and the coarse space map $\pi\colon \mathcal{X} \to C$ forgets these stabilizers. If $x_1,\dots,x_s$ are the stacky points, $e_i = \#G(x_i)$, $P_i = \pi(x_i)$, and $g = g(C)$, the *signature* of $\mathcal{X}$ is $(g; e_1,\dots,e_s)$. Following [133], the *canonical divisor* of $\mathcal{X}$ is

$$K_{\mathcal{X}} = \pi^* K_C + \sum_i (e_i - 1)\, x_i = \pi^* D \qquad \text{where} \qquad D = K_C + \sum_i \left(1 - \tfrac{1}{e_i}\right) P_i \tag{2}$$

and its degree is $\deg(K_{\mathcal{X}}) = 2g - 2 + \sum_i (1 - 1/e_i)$, which may be a rational number. Thus the genus of $\mathcal{X}$, defined by $\deg(K_{\mathcal{X}}) = 2g_{\mathcal{X}} - 2$, is the rational number $g_{\mathcal{X}} = g + \frac{1}{2}\sum_i (1 - 1/e_i) \geq g$. We say $\mathcal{X}$ is *hyperbolic* when $\deg(K_{\mathcal{X}}) > 0$, which we assume going forward. The *canonical ring* of $\mathcal{X}$ is

$$R(\mathcal{X}, K_{\mathcal{X}}) \coloneqq \bigoplus_{d\in\mathbb{Z}} H^0(\mathcal{X}, dK_{\mathcal{X}}) \cong \bigoplus_{d\in\mathbb{Z}} H^0(C, \pi_*(dK_{\mathcal{X}})) \cong \bigoplus_{d\in\mathbb{Z}} H^0(C, \lfloor dD \rfloor),$$

where the *round-down* of a $\mathbb{Q}$-divisor $D = \sum_i a_i P_i$ on C is the honest divisor $\lfloor D \rfloor \coloneqq \sum_i \lfloor a_i \rfloor P_i$, and the last isomorphism follows from [133, § 5.4]. When $\mathcal{X}$ is understood, we write R for $R(\mathcal{X}, K_{\mathcal{X}})$.

This round-down is the essential difference from the classical theory. Since $\lfloor D \rfloor = K_C$, $R_1 = H^0(\mathcal{X}, K_{\mathcal{X}}) \cong H^0(C, K_C)$ has dimension $g(C)$ rather than $g_{\mathcal{X}}$. When stacky points are present, these sections vanish along the ramification divisor $K_{\mathcal{X}} - \pi^* K_C$, so they neither embed $\mathcal{X}$ nor generate R. Adding higher-degree generators yields an embedding into a weighted projective stack [26]; algebraically, R is a quotient of a polynomial ring with a nonstandard $\mathbb{Z}$-grading.

Following Brown and Erman [34, § 3.1], define the *minimal weighted canonical series* $\mathcal{W} = \bigoplus_{d \geq 1} \mathcal{W}_d$ by setting $\mathcal{W}_1 = R_1$ and, for $d \geq 2$, choosing $\mathcal{W}_d$ as a complement in R_d to the degree-d part of the subalgebra generated by $\mathcal{W}_1, \ldots, \mathcal{W}_{d-1}$. Then $\mathcal{W}$ embeds $\mathcal{X}$ into $\mathcal{P}(\mathcal{W})$, and $R \cong S/I_{\mathcal{X}}$, where $S = \mathrm{Sym}(\mathcal{W})$. For $g \geq 2$, the stacky Noether–Petri theorem gives sharp degree bounds of $e := \max(1, e_1, \ldots, e_s)$ for generators and $2e$ for relations, subject to exceptional cases [133, Cor. 1.4.2]. Their inductive presentation also describes relative generators and gives criteria for minimality [133, Thm. 8.3.1(a),(d)]. Theorem 2.1 makes the full weight multiset explicit in our setting. Let $\nu_d = \dim \mathcal{W}_d$; the *weight multiset* of $\mathcal{W}$ has ν_d copies of each weight d, written d^{ν_d}. Set $N = \sum_d \nu_d$, $\sigma = \sum_d d\,\nu_d$, and $c = N - 2$, so $\mathcal{P}(\mathcal{W}) = \mathcal{P}(1^{\nu_1}, 2^{\nu_2}, \ldots)$ has dimension $N - 1$.

Theorem 2.1. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. The weight multiset of $\mathcal{W}$ is $(1^g,\ (2, 3, \ldots, e_1),\ \ldots,\ (2, 3, \ldots, e_s))$, with 2^{g-2} inserted if and only if C is hyperelliptic.*

For a fixed signature, the weights detect whether C is hyperelliptic; the generation degree is e for non-hyperelliptic C and $\max\{2, e\}$ otherwise. Riemann–Roch gives sections $y_{i,j} \in R_j$ for $2 \leq j \leq e_i$ with exactly one pole, at P_i of order $j - 1$. As every generator of degree m has pole order at most $m - 1$, these are indecomposable, and products $\omega \prod_s y_{i,j_s}$ with $\omega(P_i) \neq 0$ realize every admissible pole order. Filter R by pole order at the stacky points: $F_\ell R_d := H^0\big(C, dK_C + \sum_i \min\{\ell, b_i(d)\} P_i\big)$ with $b_i(d) = \lfloor d(e_i - 1)/e_i \rfloor$. This multiplicative exhaustive filtration has $F_0 R = R(C, K_C)$. Let $\Gamma_e = \{(n, \ell) \in \mathbb{N}^2 : 0 \leq \ell \leq \lfloor n \frac{e-1}{e} \rfloor\}$ and $T_e = \mathbb{C}[\Gamma_e]$, graded by n. The flat Rees degeneration $R \rightsquigarrow \mathrm{gr}_F R$ separates principal parts at the stacky points.

Theorem 2.2. *For $g \geq 3$, $\mathrm{gr}_F R$ is the fiber product of $R(C, K_C)$ and $\prod_i T_{e_i}$ over $\prod_i \mathbb{C}[z_i]$, via evaluation in fixed trivializations of $K_C|_{P_i}$ and projection to pole order zero;* $\deg z_i = 1$. *Thus there is an exact sequence*

$$0 \longrightarrow \mathrm{gr}_F R \longrightarrow R(C, K_C) \oplus \bigoplus_i T_{e_i} \longrightarrow \bigoplus_i \mathbb{C}[z_i] \longrightarrow 0. \tag{3}$$

Since $H^1(C, nK_C) = 0$ for $n \geq 2$, graded pieces are leading coefficients of principal parts, which multiply monomially at one point and vanish across distinct points. As the generators of Theorem 2.1 generate $\mathrm{gr}_F R$, $\beta(R)$ is $\beta(\mathrm{gr}_F R)$, computed from (3), minus consecutive cancellations.

2.2 The Shape of Stacky Green's Conjecture Write $K_{p,q} := \mathrm{Tor}_p^S(R, \mathbb{C})_{p+q}$ and $\beta_{p,p+q} = \dim K_{p,q}$ over the minimal ambient ring $S = \mathrm{Sym}(\mathcal{W})$. A non-hyperelliptic canonical curve has four Betti rows and $g - 1$ columns. The endpoint rows each contain one 1; duality exchanges the middle rows: $\beta_{p,j}(C; K_C) = \beta_{g-2-p,\, g+1-j}(C; K_C)$. Green's conjecture determines the linear-strand length and, with the Hilbert function, the full table for general C, whose middle rows share no internal degree. Stacky canonical rings remain Gorenstein but can have many more rows.

Theorem 2.3. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. Let $R = S/I$ be a minimal graded presentation, where $S = \mathbb{K}[x_1, \ldots, x_N]$ has total weight σ, and let $c = N - 2$.*

(1) R is Gorenstein, with $\beta_{i,j}(R) = \beta_{c-i,\sigma+1-j}(R)$.

(2) The weighted regularity, in the sense of [34]*, of R is* 3.

(3) Its nonzero Betti entries lie in rows $0 \leq j - i \leq \rho - 1$, where $\rho = \sigma - N + 4$, and both endpoint rows are attained. By Theorem 2.1, $\rho = 4 + \sum_i \binom{e_i}{2}$ if C is non-hyperelliptic and $\rho = g + 2 + \sum_i \binom{e_i}{2}$ if C is hyperelliptic.

Parts (1) and (2) follow from a local cohomology calculation showing that R is a two-dimensional Cohen–Macaulay normal domain with $\mathrm{pd}_S R = c$, together with graded local duality, or alternatively

coarse

	0	1	2	3	4
0	1	–	–	–	–
1	–	6	8	3	–
2	–	3	8	6	–
3	–	–	–	–	1

$(6;2)$

	0	1	2	3	4	5
0	1	–	–	–	–	–
1	–	6	8	3	–	–
2	–	8	24	24	8	–
3	–	–	3	8	6	–
4	–	–	–	–	–	1

$(6;3)$

	0	1	2	3	4	5	6
0	1	–	–	–	–	–	–
1	–	6	8	3	–	–	–
2	–	8	24	24	8	–	–
3	–	6	24	36	24	6	–
4	–	–	8	24	24	8	–
5	–	–	–	3	8	6	–
6	–	–	–	–	–	–	1

FIGURE 2. Betti tables of a genus-6 trigonal canonical curve C and of stacky curves $(6;2)$, $(6;3)$ over C.

from Watanabe's canonical module formula [136]. The pattern behind part (3) is striking: over a non-hyperelliptic coarse curve, whose canonical Betti table has 4 rows and $g-1$ columns, each stacky point of order e adds $e-1$ columns and $\binom{e}{2}$ rows (Figure 2). The two consequences of classical Green's conjecture thus become separate, new problems in the stacky world.

Research Problem 2.4. *Let $\mathcal{X}$ be hyperbolic, with non-hyperelliptic coarse space C of genus $g \geq 3$.*

(1) (Weak Stacky Green's Conjecture): Is a weighted analogue of Green's N_p-condition determined by $\mathrm{Cliff}(C)$ *and the signature?*

(2) (Strong Stacky Green's Conjecture): For a sufficiently general stacky curve $\mathcal{X}$, is $\beta(\mathcal{X}; K_{\mathcal{X}})$ determined by $\mathrm{Cliff}(C)$ *and its Hilbert function?*

2.3 Toward a Weak Form of Stacky Green's Conjecture Here and in §2.4, $\mathcal{X}$ and C are as in Research Problem 2.4. The weak form uses Brown–Erman's nonstandard N_p-conditions: $Z \subset \mathbb{P}(\mathcal{W})$ is *weighted* N_p if it is normally generated and F_i is generated in degree at most w^{i+1} for all $i \leq p$, where w^j sums the j largest weights [34, Def. 1.2]; standard weights recover Green's classical N_p. Let $\tau_{\mathrm{BE}}(\mathcal{X})$ be the smallest $p \geq 1$ at which weighted N_p fails. For non-stacky curves, their theorem gives weighted N_p for $1 \leq p \leq \dim \mathcal{W} - g - 2$ for any normally generated weighted series $\mathcal{W}$ of a line bundle of degree $\geq 2g+1$ with $\nu_1 > g$ [34, Thm. 1.7]. The stacky canonical series has $\nu_1 = g$, beyond their methods; [34, Ques. 7.7] requests this canonical analogue. We predict that stacky points shift the coarse threshold by $\kappa := N - g = \sum_i (e_i - 1)$, the number of higher-weight generators. Let $\tau(C, K_C)$ be the first failure of classical N_p. By duality, $\tau(C, K_C) = g - 2 - \ell(C)$, where $\ell(C) := \max\big(\{0\} \cup \{r \geq 1 : \beta_{r,r+1}(C; K_C) \neq 0\}\big)$.

Conjecture 2.5 (Weak Stacky Green's Conjecture)**.** *The threshold is $\tau_{\mathrm{BE}}(\mathcal{X}) = \tau(C, K_C) + \kappa$; equivalently, $\tau_{\mathrm{BE}}(\mathcal{X}) = (N-2) - \ell(C)$.*

For general coarse curves, Voisin's theorem [134, 135] gives $\tau(C, K_C) = \mathrm{Cliff}(C)$, so Conjecture 2.5 predicts $\tau_{\mathrm{BE}}(\mathcal{X}) = N - g + \mathrm{Cliff}(C)$, uniformly over signatures and markings. The same shift motivates Conjectures 2.13 and 2.14. The first ingredient is the following comparison of linear strands.

Proposition 2.6. *For every $r \geq 1$, the inclusion $R(C, K_C) \hookrightarrow R$ induces an isomorphism $K_{r,1}(C, K_C) \xrightarrow{\sim} K_{r,1}(\mathcal{X}, K_{\mathcal{X}})$. In particular, $\beta_{r,r+1}(\mathcal{X}; K_{\mathcal{X}}) = \beta_{r,r+1}(C; K_C)$.*

Proposition 2.6 follows by comparing Koszul cycles and boundaries using minimality of the weighted generators. Together with duality (and the final Betti entry when $\ell(C) = 0$), it gives the nonvanishing half of Conjecture 2.5. For vanishing, let u_k be the sum of the k smallest weights. Duality reduces the problem to $\beta_{r,q}(R) = 0$ for $\ell(C) < r < c$ and $q \leq u_{r+1}$. Comparison and minimality settle $r < g$. I will begin with one order-four point, where the only new requirement is $\beta_{g,g+2} = 0$, equivalently $\beta_{1,8} = 0$. The mapping cone of (3), using the semigroup relations of T_4, will test this on the special fiber and isolate any cancellation needed after deformation. For arbitrary signatures, I will insert the weighted Eagon–Northcott resolutions of the regraded rational-normal-curve rings T_{e_i} (see [64, 12]) into the fiber-product mapping cone and bound its

surviving shifts. Showing that every surviving shift in homological degree $g \le r < c$ exceeds u_{r+1} would complete the vanishing argument, proving Conjecture 2.5.

Finally, gonality and Clifford index supply no additional stacky invariant (here $\mathrm{Cliff}(\mathcal{X})$ is defined using line bundles with $h^0, h^1 \ge 2$, with the usual low-genus conventions).

Lemma 2.7 (Brill–Noether rigidity)**.** *We have* $\mathrm{Cliff}(\mathcal{X}) = \mathrm{Cliff}(C)$ *and* $\mathrm{gon}(\mathcal{X}) = \mathrm{gon}(C)$*. Every line bundle attaining either minimum is a pullback.*

Indeed, write $\mathcal{L} = \pi^* M \otimes \bigotimes_i \mathcal{T}_i^{a_i}$, where $\mathcal{T}_i = \mathcal{O}_{\mathcal{X}}(x_i)$ and $0 \le a_i < e_i$. The round-down formula gives $\dim H^i(\mathcal{X}, \mathcal{L}) = \dim H^i(C, M)$ and $\deg_{\mathcal{X}} \mathcal{L} = \deg M + \sum_i \frac{a_i}{e_i}$. Nonzero fractional parts increase degree without changing cohomology. This proves both equalities and forces every minimizer to have trivial stacky characters. The remaining Betti rows may require finer invariants.

2.4 Toward a Strong Form of Stacky Green's Conjecture The strong form asks whether g, $\mathrm{Cliff}(C)$, and the signature determine the full Betti table for general coarse C.

Theorem 2.8 (One stacky point of order two)**.** *For* $\mathcal{X} = \sqrt[2]{(C, P)}$*, with out-of-range terms zero,*

$$\beta_{p,p+q}(R) = \beta_{p,p+q}\big(R(C, K_C)\big) + \beta_{p-1,p+q-2}\big(R(C, K_C)\big) + \delta_{q,2}\tbinom{g-1}{p} - \varepsilon_{p,q},$$

where $\varepsilon_{p,q} = 1$ *exactly for* $(p, q) \in \{(1, 1), (0, 2), (g-1, 2), (g-2, 3)\}$ *and zero otherwise.*

Thus Problem 2.4(2) has an affirmative answer here: $\beta(R)$ is independent of P. Indeed, $\mathcal{W}$ has weights $1^g, 2$, so $N = g + 1$, $c = g - 1$, and $\rho = 5$. Proposition 2.6 and Theorem 2.3 specify rows $0, 1, 3, 4$, leaving only row 2. In each degree j the Hilbert numerator, which depends only on the signature, determines the sole unknown $\beta_{j-2,j}$; for $\rho \ge 6$ this fails.

Conjecture 2.9 (Strong Stacky Green's Conjecture I)**.** *The Betti numbers* $\beta_{p,q}(\mathcal{X}; K_{\mathcal{X}})$ *depend only on* C *and the signature of* $\mathcal{X}$*, and not on the position of the stacky points.*

Proposition 2.6 and Theorem 2.3 confine positional dependence to the middle rows. I will test it through the connecting-map ranks below; rank drops would identify pointed invariants beyond the gonality and Clifford index fixed by Lemma 2.7.

Our degeneration and computations suggest the following formula. For a graded S-module M, write $B_M(t, u) = \sum_{p,j} \beta^S_{p,j}(M) t^p u^j$ and $B_{\mathcal{X}} = B_R$. Let $J \subset S$ be generated by the variables of degree ≥ 2, and let P_C be the Betti polynomial of the coarse canonical ring over S/J. Set $B_J^\vee(t, u) = t^{c-1} u^{\sigma+1} B_J(t^{-1}, u^{-1})$.

Conjecture 2.10 (Strong Stacky Green's Conjecture II)**.** *We have*

$$B_{\mathcal{X}} = B_{S/J}\left(P_C + (1+tu)^{g-1} \sum_i \frac{u^2 + \cdots + u^{e_i}}{1 + tu^{e_i}}\right) - (1+t)(B_J + B_J^\vee).$$

The first term combines the coarse resolution with the higher-weight generators and the Koszul complexes on canonical forms vanishing at each P_i. The subtracted terms record consecutive cancellations paired by Gorenstein duality. To approach this formula, I will first compute the resolution of the special fiber (3), then control its cancellations under deformation.

Research Goal 2.11. *Prove that* $B_{\mathrm{gr}_F R}$ *equals the right-hand side of Conjecture 2.10 plus* $(1+t)B_J^\vee$*.*

Semicontinuity and Gorenstein duality would then bound $\beta(R)$ by the conjectured table when $(1+t)B_J$ and $(1+t)B_J^\vee$ have disjoint support, as for $\kappa < g - 2$. In this range, proving equality reduces to excluding further cancellations under deformation. To control these cancellations geometrically, M. Banks, P. Wei, and I have developed a weighted analogue of Lazarsfeld's kernel-bundle method [108, 86], as suggested in [34, §7.3]. Set $\mathcal{L} = \mathcal{O}_{\mathcal{X}}(K_{\mathcal{X}})$ and define

$$0 \longrightarrow \mathcal{M}_{\mathcal{W}} \longrightarrow \mathcal{E}_{\mathcal{W}} \coloneqq \bigoplus_{d \ge 1} \mathcal{W}_d \otimes \mathcal{L}^{-d} \longrightarrow \mathcal{O}_{\mathcal{X}} \longrightarrow 0.$$

Exterior powers and cohomology give $\operatorname{Tor}_p^S(R,\mathbb{C})_j \cong \ker H^1\big(\wedge^{p+1}\mathcal{M}_{\mathcal{W}}\otimes\mathcal{L}^j \to \wedge^{p+1}\mathcal{E}_{\mathcal{W}}\otimes\mathcal{L}^j\big)$. I will begin with $\mathcal{X}=\sqrt[3]{(C,P)}$, the first signature beyond the order-two case, with weights $1^g,2,3$. Writing $\mathcal{T}=\mathcal{O}_{\mathcal{X}}(x)$, successive evaluation maps filter $\mathcal{M}_{\mathcal{W}}$ with factors $\pi^*M_{K_C}\otimes\mathcal{L}^{-1}$, $\mathcal{L}^{-2}\otimes\mathcal{T}^{-1}$, and $\mathcal{L}^{-3}\otimes\mathcal{T}^{-1}$. Exterior powers and pushforward reduce the calculation to twists of $\wedge^a M_{K_C}$ by powers of K_C and multiples of P. The two extensions defining the filtration need not split; I will compute their connecting maps and combine them with the map to $\wedge^{p+1}\mathcal{E}_{\mathcal{W}}$ in the Tor formula. Here only rows $2,3,4$ remain undetermined, and duality exchanges rows 2 and 4, leaving $\beta_{p,p+2}$ and $\beta_{p,p+3}$ as the independent targets. For arbitrary signatures, I will express the differentials governing cancellation under the pole degeneration through the evaluation filtration and determine their ranks as the markings vary. Proving the prescribed ranks and ruling out additional cancellations would establish the full Betti formula for general markings; the loci where these ranks change would then show whether the formula extends to every marking or requires additional pointed invariants.

Schreyer's early work on Green's conjecture [125] remains, to my mind, a lovely demonstration of computation laying bare the geometry of syzygies. In that spirit, I propose to push further the computational tools that led to the conjectures and partial results above.

Research Goal 2.12. *Determine the generic Betti tables for signatures* $(g;3)$*,* $(g;4)$*, and* $(g;2,2)$ *with* $3\le g\le 6$*, letting computation test and guide Conjectures 2.9 and 2.10.*

2.5 Beyond Stacky Canonical Curves The canonical case suggests a broader principle: coarse geometry controls syzygetic thresholds, while stacky points shift their homological position. I will test this for general section rings, addressing Brown and Erman's question on weighted N_p [34, Ques. 7.6]. Let $\mathcal{D}=\pi^*L\otimes\bigotimes_i\mathcal{T}_i^{a_i}$, where L is very ample on C, $0<a_i<e_i$, $\gcd(a_i,e_i)=1$, and $\mathcal{T}_i=\mathcal{O}_{\mathcal{X}}(x_i)$ as above, so $\deg\mathcal{T}_i=1/e_i$, $\mathcal{T}_i^{e_i}\cong\pi^*\mathcal{O}_C(P_i)$, and $\lfloor\mathcal{D}\rfloor=L$. When $\deg L\ge 2g-1$, the coarse H^1 vanishings extend the pole filtration of Theorem 2.2 to $R(\mathcal{X},\mathcal{D})$, replacing $b_i(d)$ by $\lfloor da_i/e_i\rfloor$ and T_{e_i} by the semigroup ring of $\{(n,\ell):0\le\ell\le\lfloor na_i/e_i\rfloor\}$.

The first regime is $\deg L\ge 2g+1$, where classical vanishing theorems [87, 86] provide a starting point. Let $\mathcal{W}$ minimally generate $R(\mathcal{X},\mathcal{D})$ and set $\kappa=\dim\mathcal{W}-h^0(C,L)$, the number of generators of weight greater than one. This depends on the stacky twist: $\kappa=\sum_i(e_i-1)$ when every $a_i=e_i-1$, but $\kappa=s$ when every $a_i=1$. I predict that these generators shift the coarse threshold without otherwise changing it. As in §2.3, $\tau_{\mathrm{BE}}(\mathcal{X},\mathcal{D})$ and $\tau(C,L)$ denote the first failures of weighted and ordinary N_p, respectively, with either threshold equal to ∞ if no failure occurs.

Conjecture 2.13. *If* $\deg L\ge 2g+1$*, then* $\tau_{\mathrm{BE}}(\mathcal{X},\mathcal{D})=\tau(C,L)+\kappa$.

I will begin with one stacky point of order e and $\mathcal{D}=\pi^*L\otimes\mathcal{T}$, so $\kappa=1$. Write $R=R(\mathcal{X},\mathcal{D})$. The evaluation filtration has factors $\pi^*M_L\otimes\mathcal{D}^{-1}$ and $\mathcal{D}^{-e}\otimes\mathcal{T}^{-1}$. In the first forbidden degree $j=p+e+1$, exterior powers and pushforward give $\wedge^{p+1}M_L\otimes L^e(P)$ and $\wedge^pM_L\otimes L$. Coarse regularity kills the first H^1, the target H^1 in the Tor formula, and the terms in higher forbidden degrees; the second is $K_{p-1,2}(C,L)$. The first target is therefore the comparison $\operatorname{Tor}_p^S(R,\mathbb{C})_{p+e+1}\cong K_{p-1,2}(C,L)$, proving the shift by one. For general κ, I will track the vanishing orders removed by successive evaluation maps and show that only the factor containing all higher-weight generators can contribute in forbidden degrees. This would recover the coarse Koszul group and explain the full shift. I will then test the method on log canonical rings, including modular-form rings with cusps, without using Gorenstein duality. This complements recent work in nonstandard gradings [32, 31, 11, 114, 63, 37].

At degree $2g$, secant geometry gives a sharper target. For $g=2n+1$ and $\operatorname{Cliff}(C)=n$, Farkas–Kemeny show that a very ample, normally generated line bundle L of degree $2g$ satisfies N_{n-1} if and only if it is *n-very ample*: every subscheme of length $n+1$ imposes independent conditions [74, Thm. 1.4]; cf. [73]. I predict that this criterion, shifted by κ, persists for general stacky points.

Conjecture 2.14 (Stacky Secant Conjecture)**.** *For C and L as above, fixed $e_i\ge 2$, and general distinct $P_i\in C$, set $\mathcal{D}=\pi^*L\otimes\bigotimes_i\mathcal{T}_i^{e_i-1}$, $\kappa=\sum_i(e_i-1)$. Then $(\mathcal{X},\mathcal{D})$ satifies weighted $N_{n-1+\kappa}$ if and only if L is n-very ample.*

The first test is genus five with one point of order two: weighted N_2 should distinguish general degree-ten line bundles from those with a length-three secant obstruction. Taking $L = K_C(D - Q)$, with D a general effective divisor of degree three and $Q \notin \operatorname{Supp}(D)$, gives an explicit family: L is nonspecial, but D imposes only two conditions. For arbitrary odd genus and signature, I will filter the evaluation kernel one stacky generator at a time and seek a comparison of all groups forbidden by weighted $N_{n-1+\kappa}$ with shifted coarse Koszul groups. The key is to use the 2-regularity of $R(C, L)$ to eliminate the other H^1-terms, leaving Farkas–Kemeny's theorem to identify the surviving coarse obstruction with failure of n-very ampleness.

3. Multigraded Castelnuovo–Mumford Regularity: Bounds and Behavior

The PI linked cohomological vanishing and syzygies on products of projective spaces and proved finite-generation and asymptotic results for toric regularity (§1, [40, 41]). Yet regularity can have infinitely many corners, multigraded Betti numbers do not determine it, and geometric bounds remain scarce. I will explain its shape through support geometry, bound corners using syzygies, and bound curve regularity through intersection theory. Throughout, X, S, B, $B_1, \ldots, B_s$ are as in §1, $n = \dim X$, and $\mathbb{K}$ is algebraically closed of characteristic zero.

3.1 The Shape and Minimal Degrees of Multigraded Regularity As the PI showed, the coordinate ring S/I_p of a point $p \in \mathcal{H}_2$ has infinitely many minimal regular degrees (Figure 1(c), [40]), although $\mathcal{O}_p$ is regular in every degree. These corners are measured relative to $\operatorname{Nef}(X)$, and I propose that their infinitude reflects the choice of cone: the ambient nef cone can be too small to account for the directions of positivity available to M. A cone adapted to M should recover persistence and finite generation, up to a uniform shift. The point example supports this expectation: its support-relative regularity has only one corner. This leads to three connected problems: determine the directions of regularity from geometry, bound its minimal degrees using syzygies, and control the number of corners. For a finitely generated graded S-module M, set $\mathcal{F} = \widetilde{M} \neq 0$ and $Z = V(\operatorname{ann} \mathcal{F})$. With D_ρ the torus-invariant prime divisor of a ray ρ, set

$$K_Z = \bigcap_{p \in Z} \operatorname{cone}\{[D_\rho] : p \notin D_\rho\}, \qquad C_Z = \overline{\operatorname{cone}\{[C] : C \subset Z \text{ an integral curve}\}}^{\vee},$$

with curve classes in $N_1(X)_{\mathbb{R}}$. A rational class lies in K_Z if some positive multiple has no ambient base point on Z, and lies in C_Z if it is nef on Z; thus $\operatorname{Nef}(X)_{\mathbb{R}} \subset K_Z \subset C_Z$. The cone K_Z records which torus orbits meet Z, while C_Z records its curve geometry. I conjecture that these are the recession cones of the closed convex hulls of $\operatorname{reg}(M)$ and $\operatorname{reg}(\mathcal{F})$.

Conjecture 3.1. *If $H^0_B(M) = 0$, there are $\mathbf{a}_M, \mathbf{b}_M, \mathbf{a}_\mathcal{F}, \mathbf{b}_\mathcal{F} \in \operatorname{Pic}(X)$ such that*

$$\mathbf{a}_M + (K_Z \cap \operatorname{Pic}(X)) \subset \operatorname{reg}(M) \subset \mathbf{b}_M + K_Z \quad \textit{and} \quad \mathbf{a}_\mathcal{F} + (C_Z \cap \operatorname{Pic}(X)) \subset \operatorname{reg}(\mathcal{F}) \subset \mathbf{b}_\mathcal{F} + C_Z.$$

If $H^0_B(M) = 0$, only the lifting obstruction $H^1_B(M)$ separates module and sheaf regularity. In Figure 1(c), $\operatorname{reg}(S/I_p)$ fills $K_p \supsetneq \operatorname{Nef}(X)_{\mathbb{R}}$, while $C_p = \operatorname{Pic}(X)_{\mathbb{R}}$. For the diagonal $\Delta \subset \mathbb{P}^1 \times \mathbb{P}^1$, K_Δ is the nef cone but C_Δ is a half-plane: $\operatorname{reg}(S/(x_0y_1 - x_1y_0)) = \mathbb{N}^2$, while $\operatorname{reg}(\mathcal{O}_\Delta) = \{(a, b) : a + b \geq 0\}$, so one module corner shatters into infinitely many sheaf corners. I will approach the module bounds through toric modifications of linear systems whose base loci avoid Z. For sheaves, Fujita vanishing on Z [99, Thm. 1.5] and the global generation forced by regularity point to the two inclusions, and comparing first syzygies along support curves offers a route to effective translates [111].

Our faithfulness theorem (§1) gives finitely many corners but does not locate them, and when $K_Z \supsetneq \operatorname{Nef}(X)_{\mathbb{R}}$, Conjecture 3.1 forces infinitely many. I therefore measure corners relative to $\Gamma_Z \coloneqq K_Z \cap \operatorname{Pic}(X)$: let $\operatorname{reg}_Z(M)$ be Maclagan–Smith regularity [111] with respect to $B_1, \ldots, B_s$ together with the minimal generators of Γ_Z, so that Γ_Z plays the role of $\operatorname{Nef}(X)$. Then $\operatorname{reg}_Z(M) \subseteq \operatorname{reg}(M)$ is Γ_Z-stable, with equality on products, and in Figure 1(c), $\operatorname{reg}_Z(S/I_p) = \Gamma_p$ has a single corner. I conjecture that, up to a uniform shift, regularity *is* support-relative regularity, with corners in a box determined by the Betti table.

Conjecture 3.2. *Fix X and a basis of* $\mathrm{Pic}(X)$*. There are* $\mathbf{c}_X \in \mathrm{Nef}(X)$ *and, for each multigraded Betti table* β*, an explicit finite box* $Q_X(\beta)$ *such that every finitely generated graded S-module M with* $\beta^S(M) = \beta$*,* $\widetilde{M} \neq 0$*, and* $H^0_B(M) = 0$ *satisfies*

$$\mathbf{c}_X + \mathrm{reg}(M) \ \subseteq \ \mathrm{reg}_Z(M) \ = \ \big(Q_X(\beta) \cap \mathrm{reg}_Z(M)\big) + \Gamma_Z.$$

The inclusion gives persistence along Γ_Z, and the equality locates its corners; together they imply the module half of Conjecture 3.1. I will approach the inclusion by adapting the Maclagan–Smith nonzerodivisor argument to sufficiently divisible degrees in Γ_Z, tracking the nef shift through induction on $\dim Z$. For the box, Lauren Cranton Heller and I have proved the following:

Lemma 3.3. *Under the hypotheses of Conjecture 3.2, with* $X = \mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$*, every corner of* $\mathrm{reg}(M)$ *lies in* $\prod_{j=1}^{2}[\ell_j(\beta), U_j(\beta)]$*, where* $\ell_j(\beta) = \min_{\beta_{0,\mathbf{a}} \neq 0} a_j$ *and* $U_j(\beta) = \max_{\beta_{i,\mathbf{a}} \neq 0} a_j$*.*

For $r \geq 3$, I will seek coordinatewise descent: if $\mathbf{d} \in \mathrm{reg}(M)$ and $d_j > U_j(\beta)$, then $\mathbf{d} - \mathbf{e}_j \in \mathrm{reg}(M)$; iterating would trap corners in a box. I will start with three factors, isolating one direction via partial Koszul complexes, then extend to faces of Γ_Z via diagonal resolutions [25, 50, 94, 36].

Corollary 3.4. *Assume Conjecture 3.2 and fix* β*. Among modules satisfying its hypotheses, only finitely many regions* $\mathrm{reg}_Z(M)$ *occur. Each is determined by* Γ_Z *and the finite set* $Q_X(\beta) \cap \mathrm{reg}_Z(M)$*, and* $\mathrm{reg}_Z(M) \subseteq \mathrm{reg}(M) \subseteq \mathrm{reg}_Z(M) - \mathbf{c}_X$*.*

This holds as Γ_Z depends only on which torus orbits meet Z, so only finitely many monoids occur; testing a corner still needs vanishing on all of Γ_Z. On products, $\Gamma_Z = \mathrm{Nef}(X)$ and the Maclagan–Smith criterion applies, so the $r = 2$ box theorem gives a complete algorithm for $\mathrm{reg}(M)$ on $\mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$, towards [111, Prob. 4.10]; existing implementations, including ours [3], scan a fixed window and can miss corners outside it. In general, I will seek a finite criterion adapting the nonzerodivisor argument of [111, Cor. 4.8]. Our asymptotic theorem traps $\mathrm{reg}(I^p)$ between translates moving linearly in p, but corners may run superlinearly along faces.

Conjecture 3.5. *Let* $X = \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_r}$*,* $I \subset S$ *be homogeneous, and M be a finitely generated graded S-module. There are constants* A, C *such that* $\|\mathbf{v}\|_\infty \leq Ap + C$ *for every corner* $\mathbf{v} \in \mathrm{reg}(I^p M)$*.*

The Betti degrees of $I^p M$ are bounded linearly in p [9, Thm. 1.1], so the proposed box on products would imply this when $H^0_B(M) = 0$; our $r = 2$ theorem already gives this case. For general M, I will separately bound the contribution of B-torsion using the Rees module. Since corners form an antichain, the window allows $O(p^{r-1})$ of them, and this is sharp: for the ideal $\mathfrak{m}$ of all Cox variables, $\mathrm{reg}(\mathfrak{m}^p) = \{\mathbf{d} \in \mathbb{N}^r : |\mathbf{d}| \geq p\}$ has $\binom{p+r-1}{r-1}$ corners. For ordinary powers I predict an arithmetic structure. Set $\mathcal{C}(I) \coloneqq \{(p, \mathbf{d}) \in \mathbb{N} \times \mathbb{Z}^r : p \geq 1, \ \mathbf{d} \text{ is a corner of } \mathrm{reg}(I^p)\}$, and call a set *semilinear* if it is a finite union of translates of finitely generated additive monoids.

Conjecture 3.6. *Let* $X = \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_r}$ *with* $n_j \geq 1$*, and let* $0 \neq I \subsetneq S$ *be homogeneous. There is* p_0 *such that* $\mathcal{C}(I) \cap \{p \geq p_0\}$ *is semilinear.*

This would make the number of corners eventually quasipolynomial in p, of degree at most $r - 1$ [137]. I will begin with monomial ideals, adapting constructibility under local cohomology [62, Thm. 8.1] and using Presburger quantification over fine degrees and nef shifts to describe regularity and its corners. In general, I will test Rees degenerations on $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \mathbb{P}^2$, seeking conditions that prevent cohomological cancellation along the boundary. Even where corners change, this gives explicit bounds complementing [42].

3.2 A Bound for Multigraded Regularity in the Style of Gruson, Lazarsfeld, and Peskine A foundational result of Gruson, Lazarsfeld, and Peskine [90] states that if $C \subset \mathbb{P}^r_{\mathbb{K}}$ is an integral nondegenerate curve defined over an algebraically closed field $\mathbb{K}$, then $\mathrm{reg}\,(I_C) \leq \deg(C) - r + 2$, where $I_C \subset \mathbb{K}[x_0, \ldots, x_r]$ is the (saturated) homogeneous defining ideal of C. Since $\deg(C) = C \cdot H$ for the hyperplane class H, the theorem is a strong bridge between the intersection theory of C and

the degrees of its equations and syzygies. It sharpens classical work of Castelnuovo [46] on space curves and settles the curve case of the Eisenbud–Goto conjecture $\operatorname{reg}(I_C) \leq \deg(C) - \operatorname{codim}(C) + 1$ [69]. It has inspired decades of work on optimal bounds [123, 107, 104, 105, 83] and on bounding regularity by the degrees of defining equations [19, 67, 61, 110], and recent counterexamples to the general Eisenbud–Goto conjecture [115, 47, 116] make such geometric bounds all the more valuable.

Research Problem 3.7. *Prove a multigraded analogue of Gruson, Lazarsfeld, and Peskine's (GLP) theorem for curves in smooth projective toric varieties.*

I seek an explicit region in $\operatorname{reg}(\mathcal{I}_C)$ shaped by X's divisor relations [111]. Cobb adapted the kernel bundle techniques of GLP to products of projective spaces [90, 53], and Brown–Erman ask for a weighted analogue [34, Ques. 7.5]. Classically, if $m = \deg(C) - r + 2 \geq 2$, then the exact sequence $0 \to \mathcal{I}_C \to \mathcal{O}_{\mathbb{P}^r} \to \mathcal{O}_C \to 0$ reduces regularity to three tests. (i) *Ambient* ($q \geq 3$): $H^q(\mathcal{I}_C(m-q)) \cong H^q(\mathcal{O}_{\mathbb{P}^r}(m-q)) = 0$, since $\dim C = 1$ and projective-space vanishing includes $q = r$ when $m \geq 0$. (ii) *Speciality* ($q = 2$): $H^2(\mathcal{I}_C(m-2)) \cong H^1(\mathcal{O}_C(m-2)) = 0$, since Castelnuovo's genus bound gives $(m-2)\deg(C) > 2p_a(C) - 2$. (iii) *Interpolation* ($q = 1$): $H^1(\mathcal{I}_C(m-1))$ is the cokernel of $H^0(\mathcal{O}_{\mathbb{P}^r}(m-1)) \to H^0(\mathcal{O}_C(m-1))$, vanishing exactly when sections on C extend to $\mathbb{P}^r$. This surjectivity can be proved via kernel bundle techniques. Nondegeneracy gives $\operatorname{length}(C \cap \ell) \leq \deg(C) - r + 2$ for every secant line ℓ; interpolation on an s-secant line forces $\operatorname{reg}(\mathcal{I}_C) \geq s$, so maximal secants govern the bound.

Now let X be as in this section with $n \geq 2$. Say that an integral curve $C \subset X$ is *nondegenerate* if each restriction $H^0(X, B_j) \to H^0(C, B_j|_C)$ is injective. Set $r_j := \dim_{\mathbb{K}} H^0(X, B_j) - 1$, $d_j := \langle B_j \cdot C \rangle$, and the *defect* $\delta_j := d_j - r_j$. Maclagan–Smith regularity of $\mathcal{I}_C$ at $\mathbf{d}$ tests the twists $\mathbf{d} - B_i$, then $\mathbf{d} - B_i - B_j$, and so on, so the same three questions arise. I propose answering each by a region depending only on X and the numerical class of C: (i) *Ambient.* Membership in $\operatorname{reg}(\mathcal{O}_X)$ gives $H^q(X, \mathcal{O}_X(\mathbf{d} - \sum_j v_j B_j)) = 0$ for $q > 0$ and $|\mathbf{v}| = q$. Since C is a curve, for $q \geq 3$ these are exactly the ideal-sheaf tests. (ii) *Speciality.* Choose an explicit numerical bound $\widehat{G}_X([C]) \geq p_a(C)$. For example, $A = \sum_j B_j$ is very ample, and multiplication of the restricted section spaces shows that C spans at least a $\mathbb{P}^R$, where $R = \sum_j r_j$. Thus one may take Castelnuovo's bound $\pi(D, R)$ [6, Ch. III], where $D = \sum_j d_j$: writing $D - 1 = m(R-1) + \epsilon$ with $0 \leq \epsilon < R - 1$, this is $\binom{m}{2}(R-1) + m\epsilon$. On a surface, adjunction gives the exact genus and can be used instead. Every second twist is nonspecial on $\mathcal{H}_X([C]) := \{\mathbf{d} \in \operatorname{Pic}(X) : \mathbf{d} \cdot C \geq 2\widehat{G}_X([C]) - 1 + 2\max_j d_j\}$, because $\deg \mathcal{O}_C(\mathbf{d} - B_i - B_j) > 2p_a(C) - 2$. (iii) *Interpolation.* Guided by the secant-line explanation above, set $M_X := \overline{\operatorname{NE}}(X) \cap N_1(X)_{\mathbb{Z}}$ and $e_j(\gamma) := B_j \cdot \gamma$. Call $0 \neq \gamma \in M_X$ *eligible* if $e_j(\gamma) < r_j$ for some j, and put $q_C(\gamma) := \min_{j:\, e_j(\gamma) < r_j}(\delta_j + e_j(\gamma))$ and $w(\gamma) := \max_j e_j(\gamma)$. If an integral rational curve $C' \neq C$ has eligible class γ, its image under B_j spans at most $\mathbb{P}^{e_j(\gamma)}$. The resulting sections vanishing on C' and nondegeneracy of C give $\operatorname{length}(C \cap C') \leq q_C(\gamma) + 1$. When C' is smooth, any values on a subscheme of length $q_C(\gamma) + 1$ are interpolated by line bundles of degree $\geq q_C(\gamma)$. Requiring this degree at every first twist motivates:

$$\mathcal{S}_X([C]) := \{\mathbf{d} \in \operatorname{Pic}(X) \mid \mathbf{d} \cdot \gamma \geq q_C(\gamma) + w(\gamma) \text{ for every eligible } \gamma \in M_X\}.$$

Conjecture 3.8 (Multigraded GLP)**.** *If $C \subset X$ is an integral nondegenerate curve, then*

$$\mathcal{R}_X([C]) := \mathcal{S}_X([C]) \cap \mathcal{H}_X([C]) \cap \operatorname{reg}(\mathcal{O}_X) \;\subseteq\; \operatorname{reg}(\mathcal{I}_C).$$

On $\mathbb{P}^r$, the line class gives $m \geq \deg(C) - r + 2$, higher eligible multiples give weaker inequalities, and Castelnuovo's genus bound supplies nonspeciality at the same threshold, so the conjecture recovers the classical theorem. I have proved it for curves in $\mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$ with a minimal-degree projection, and computations for curves in $\operatorname{Bl}_p \mathbb{P}^3$ have found no counterexamples. The bound is sharper than Cobb's even for simple curves. Consider the smooth image $C \subset X = \mathbb{P}^2 \times \mathbb{P}^2$ of $[u : v] \mapsto \big([u^3 : u^2v : v^3], [u^2 : uv : v^2]\big)$, with multidegree $(3, 2)$ and defects $(1, 0)$. The inequalities defining $\mathcal{S}_X([C])$ reduce to $a \geq 1$ and $b \geq 2$; the genus estimate $\widehat{G}_X([C]) = \pi(5, 4) = 1$ adds only the

redundant $3a + 2b \geq 7$. In fact, $\mathcal{R}_X([C]) = \operatorname{reg}(\mathcal{I}_C) = (1,2) + \mathbb{N}^2$, so the bound is sharp (Lozovanu also obtains this corner over $\mathbb{C}$ [109, Thm. A]), whereas Cobb's bound gives $(3,3) + \mathbb{N}^2$ [53, Prop. B].

What remains is precise: the ambient and nonspeciality conditions give the other regularity tests, so it suffices to prove that $H^0(X, \mathcal{O}_X(\mathbf{d} - B_i)) \to H^0(C, \mathcal{O}_C(\mathbf{d} - B_i))$ is surjective for all $\mathbf{d} \in \mathcal{R}_X([C])$ and i. For general X, I will combine the classical auxiliary line bundle on the normalization $f\colon \widetilde{C} \to X$ with the Hanlon–Hicks–Lazarev diagonal [90, 53, 94], whose terms are $\mathcal{D}_p = \bigoplus_\lambda \mathcal{O}_X(-A_{p\lambda}) \boxtimes \mathcal{O}_X(-E_{p\lambda})$. For a line bundle L on $\widetilde{C}$, pulling back, tensoring, and pushing forward give a spectral sequence with $E_1^{-p,q} = \bigoplus_\lambda H^q\big(\widetilde{C}, L \otimes f^*\mathcal{O}_X(-A_{p\lambda})\big) \otimes \mathcal{O}_X(-E_{p\lambda})$ for $q = 0, 1$, converging to f_*L in total degree zero. I will first use Brown–Sayrafi's diagonal on $\operatorname{Bl}_p \mathbb{P}^3$ [36]. For general L, the inequalities $\deg L - A_{p\lambda} \cdot C \geq g(\widetilde{C}) - 1$ kill the relevant H^1. I will compute the resulting presentation and test whether its determinantal complex yields first-twist surjectivity throughout $\mathcal{R}_X([C])$, beginning with smooth curves. Should extra positivity be needed, I will obtain an explicit weaker region and identify the missing sharpening. For singular curves, I will track the normalization quotient; curves in $\mathbb{P}^1 \times \mathbb{P}^2$ are a first graduate-student project.

3.3 Multigraded Regularity of Torus-Invariant Subvarieties On a smooth projective toric variety $X = X_\Sigma$, the torus-invariant irreducible subvarieties are in dimension-reversing bijection with the cones of Σ. If $S = \mathbb{K}[x_\rho \mid \rho \in \Sigma(1)]$ is the Cox ring, the subvariety Z_σ corresponding to $\sigma \in \Sigma$ has Cox ideal $I_{Z_\sigma} = \langle x_\rho \mid \rho \in \sigma(1) \rangle$, which is resolved by a multigraded Koszul complex. Even so, the multigraded regularity of these subvarieties can be subtle: the four torus-fixed points of the Hirzebruch surface $\mathcal{H}_2$ exhibit two distinct regularities, one with infinitely many corners [40]. Thus they are a natural testing ground, but a concrete description of their regularity is unknown.

Research Problem 3.9. *Let $Z_\sigma \subset X$ be an irreducible torus-invariant subvariety corresponding to the cone σ. Give an explicit combinatorial characterization of* $\operatorname{reg}(I_{Z_\sigma})$ *and* $\operatorname{reg}(S/I_{Z_\sigma})$.

We ask for both because, unlike in the standard graded case, $\operatorname{reg}(S)$ need not have a single corner, so the two can behave quite differently [40]. Our approach combines Hilbert-function combinatorics with explicit higher cohomology of line bundles on torus-invariant subvarieties. With Jacob Lehman Duke, I have settled torus-fixed points. For a maximal cone σ, the variables surviving in S/I_{Z_σ} are indexed by rays outside σ, and the degrees of surviving monomials form the monoid $Q_\sigma = \{\sum_{\rho \notin \sigma(1)} a_\rho [D_\rho] \mid a_\rho \in \mathbb{N}\} \subset \operatorname{Pic}(X)$. By smoothness, the classes $[D_\rho]$ for $\rho \notin \sigma(1)$ form a lattice basis of $\operatorname{Pic}(X)$, so the Hilbert function of S/I_{Z_σ} is 1 on Q_σ and 0 elsewhere.

Proposition 3.10. *If $Z_\sigma \subset X$ is a torus-fixed point corresponding to a maximal cone $\sigma \in \Sigma$, then* $\operatorname{reg}(S/I_{Z_\sigma}) = Q_\sigma$ *and* $\operatorname{reg}(I_{Z_\sigma}) = \operatorname{reg}(S) \cap \bigcap_{j=1}^s (B_j + Q_\sigma)$.

Since S/I_{Z_σ} has no higher sheaf cohomology, its regularity is the set of degrees where the Hilbert function and polynomial agree [111, Prop. 6.4], namely Q_σ. A cohomology argument reduces $\operatorname{reg}(I_{Z_\sigma})$ to surjectivity of evaluation at each first twist, i.e. $\mathbf{d} - B_j \in Q_\sigma$ for all j, giving the intersection. $\operatorname{reg}(S/I_{Z_\sigma})$ depends only on σ and the rays of Σ, so it is shared by all toric models with these data. Next are invariant curves $Z_\sigma \cong \mathbb{P}^1$, whose sheaf regularity is the half-space $\mathbf{d} \cdot Z_\sigma \geq \max_j(B_j \cdot Z_\sigma) - 1$, so I will determine which sections the surviving Cox monomials represent. In higher dimension, I will combine these character counts with the toric Čech complex of the star of σ.

Such descriptions would also make the degeneration bounds of Maclagan and Smith explicit: they bound the regularity of a monomial quotient by intersecting shifted regularity regions of the factors in a Stanley filtration, which for saturated monomial ideals are coordinate rings of orbit closures [112, Thm. 4.1]. Answering Problem 3.9 combinatorially would compute these from the fan, and semicontinuity carries the bounds from initial ideals to the quotient [111, Prop. 6.13].

4. Uniform Homological Algebra: Regularity on the Cox Category

I propose a deliberately speculative test: *multigraded regularity may be asking the right questions, but in the wrong category.* Work of the PI and others shows that the classical link between cohomological

vanishing and syzygies is fragile [111, 41, 40]: even the Cox ring of the Hirzebruch surface $\mathcal{H}_2$, free over itself, is not $\mathbf{0}$-regular [111, Ex. 4.3]. Uniformity across toric birational models is a growing theme. It begins with the Hanlon–Hicks–Lazarev resolution of the diagonal, which depends only on the rays of the fan [94], and extends to short resolutions, cellular constructions, and derived categories [33, 77, 78, 79, 58, 23, 17, 93, 71]. King's conjecture, that every smooth projective toric variety has a full strong exceptional collection of line bundles, is false [96, 117]; Ballard et al. recover it by passing to the *Cox category*, which sees all birational GKZ models at once [10]. I will use this uniform diagonal to define regularity where the structure sheaf is $\mathbf{0}$-regular on every smooth model.

4.1 The Cox Category and Diagonal Heights Let X be a projective normal toric variety over an algebraically closed field $\mathbb{K}$ of characteristic zero, with Cox ring S. The maximal GKZ chambers give models X_i and smooth toric stacks $\mathcal{X}_i$, with $\mathcal{X}_i = X_i$ when X_i is smooth. Following [10], choose a common smooth toric stack with proper birational maps $\pi_i\colon \widetilde{\mathcal{X}} \to \mathcal{X}_i$ and set $D_{\mathrm{Cox}}(X) := \langle \pi_i^* D^b(\mathcal{X}_i)\rangle \subseteq D^b(\widetilde{\mathcal{X}})$. Here $D^b(-) = D^b(\mathrm{Coh}(-))$, and all functors are derived. The category is independent of this choice and contains each $D^b(\mathcal{X}_i)$ fully faithfully. The divisor-class map $Q\colon \mathbb{Z}^{\Sigma(1)} \to \mathrm{Cl}(X)$ determines the half-open zonotope $Z_X = Q_{\mathbb{R}}((-1,0]^{\Sigma(1)})$ and the finite *Bondal–Thomsen collection* Θ of classes mapping into Z_X [132, 21]. Pulling back $\mathcal{O}(\alpha)$ from a chamber containing $-\alpha$ gives objects $T_\alpha := \mathcal{O}_{\mathrm{Cox}}(\alpha)$ forming a full strong exceptional collection [10].

The uniform HHL complex lifts to a diagonal kernel inducing the identity on $D_{\mathrm{Cox}}(X)$ [10, § 6]. Take its minimal representative in the additive category generated by the fixed external products $T_\alpha \boxtimes T_\beta$, with terms $D_p = \bigoplus_{\alpha,\beta\in\Theta}(T_\alpha \boxtimes T_\beta)^{m_{p,\alpha,\beta}}$ for $p \geq 0$. Use the first factor as cohomological input, with $p = 0$ adjacent to the augmentation. For the occurring input labels $\alpha \in \Theta_H$, set $h(\alpha) := \min\{p : m_{p,\alpha,\beta} \neq 0 \text{ for some } \beta\}$ and $\nu(\alpha) := \max\{1, h(\alpha)\}$. The *height* records first occurrence in the minimal diagonal, not necessarily face codimension in the zonotope.

Definition 4.1. *Call* $\mathcal{E} \in D_{\mathrm{Cox}}(X)$ $\mathbf{0}$-(Cox) regular *if* $\mathrm{Ext}^q(T_\alpha^\vee, \mathcal{E}) = 0$ *for* $\alpha \in \Theta_H$, $q \geq \nu(\alpha)$.

Degree-q cohomology in diagonal degree p contributes in position $p - q$. The cutoff leaves only positive positions at positive height and allows sections at height zero. On $\mathbb{P}^n$, $h(-j) = j$ for $0 \leq j \leq n$, recovering classical 0-regularity. Regularity relative to an exceptional collection goes back to Costa–Miró-Roig [55, 56]; the new feature is the height cutoff, uniform across models.

Research Problem 4.2. *Find a combinatorial formula for* $h(\alpha)$ *from the zonotope and GKZ fan.*

Starting from the Betti-number formulas of Favero–Sapronov and Han [79, 93], I will express Cox heights directly through zonotope geometry and GKZ wall identifications. In Figure 3, two labels of different heights become the same line bundle on the weighted projective stack.

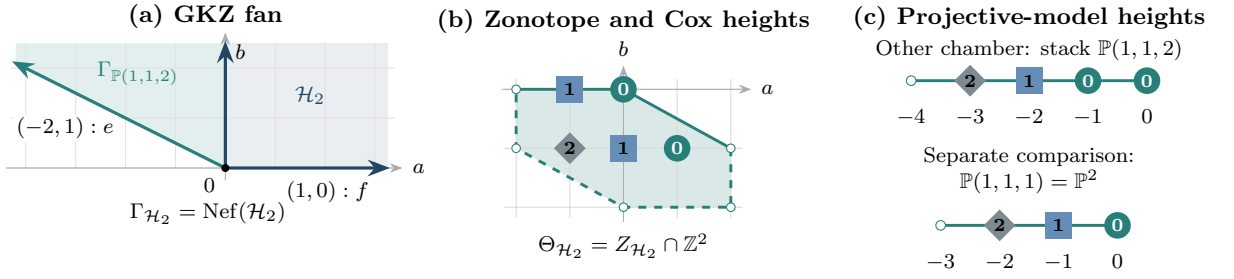


FIGURE 3. Diagonal heights for $\mathcal{H}_2$. (a) GKZ chambers; contracting e gives $\mathbb{P}(1,1,2)$. (b) Heights from the diagonal of [58, §4]; dashed edges and open vertices are excluded. (c) Heights on $\mathbb{P}(1,1,2)$ and on $\mathbb{P}^2$.

4.2 The Structure Sheaf and Nef Geometry To compare with classical regularity, fix a smooth projective model X_i; write α_i for the image of α on the ith model. For a coherent sheaf $\mathcal{F}$ on X_i, let $\mathcal{R}_i(\mathcal{F}) \subseteq \mathrm{Pic}(X_i)$ consist of the $\mathbf{d}$ for which $\pi_i^*(\mathcal{F}(\mathbf{d}))$ is $\mathbf{0}$-regular. The Θ-Transform Lemma [10, Lemma 1.6] and the projection formula give $\mathrm{Ext}^q(T_\alpha^\vee, \pi_i^*(\mathcal{F}(\mathbf{d}))) \cong H^q(X_i, \mathcal{F}(\mathbf{d} + \alpha_i))$. Thus $\mathbf{d} \in \mathcal{R}_i(\mathcal{F})$ exactly when $H^q(X_i, \mathcal{F}(\mathbf{d} + \alpha_i)) = 0$ for all $\alpha \in \Theta_H$ and $q \geq \nu(\alpha)$.

Lemma 4.3. *The object $\mathcal{O}_{\mathrm{Cox}}$ is* $\mathbf{0}$*-regular, and* $\mathrm{Nef}(X_i) \subseteq \mathcal{R}_i(\mathcal{O}_{X_i})$ *on every smooth model* X_i*.*

Bondal–Thomsen vanishing gives $H^q(\mathcal{X}_i, \mathcal{O}_{\mathcal{X}_i}(\alpha_i)) = 0$ and $H^q(X_i, \mathcal{O}_{X_i}(\mathbf{d} + \alpha_i)) = 0$ for $q > 0$, $\alpha \in \Theta$, and nef $\mathbf{d}$ [10, Thm. 2.11]. Since $\nu(\alpha) \geq 1$ and $\mathcal{O}_{\mathrm{Cox}}$ is the common pullback of the structure sheaves, both claims follow. So, unlike Maclagan–Smith regularity, Cox regularity makes the structure sheaf $\mathbf{0}$-regular on every model, as on $\mathbb{P}^n$. The lemma uses only $\nu \geq 1$; the heights enter in the converse: does Cox regularity keep the positivity consequences of classical regularity?

Question 4.4. *Let $\mathcal{F}$ be a coherent sheaf on a smooth projective model X_i. Is $\mathcal{R}_i(\mathcal{F}) + \mathrm{Nef}(X_i) \subseteq \mathcal{R}_i(\mathcal{F})$? Is $\mathcal{F}(\mathbf{d})$ globally generated for every $\mathbf{d} \in \mathcal{R}_i(\mathcal{F})$?*

Since globally generated line bundles are nef, global generation would give $\mathcal{R}_i(\mathcal{O}_{X_i}) = \mathrm{Nef}(X_i)$. To prove persistence, I will resolve $\mathcal{O}_{X_i}(\alpha_i + B_j)$, for Hilbert-basis generators B_j of $\mathrm{Nef}(X_i)$, by sums of test line bundles $\mathcal{O}_{X_i}(\beta_i)$ with $\nu(\beta) \leq \nu(\alpha) + p$ in homological degree p; the cohomology spectral sequence then gives vanishing, and iteration handles all nef translates. I will construct these certificates first on Hirzebruch surfaces, then on scrolls. For global generation, the height cutoff leaves only diagonal-degree-zero sections in position zero; I will seek to absorb their combined image into ordinary global evaluation.

The decisive test is multiplication of sections. On $\mathbb{P}^n$, Mumford's theorem makes multiplication by sections surjective above the regularity of a locally free sheaf. Maclagan–Smith regularity keeps this [111], but at the cost seen on $\mathcal{H}_2$: nef twists of $\mathcal{O}$ need not be regular. Cox regularity reverses the trade-off, making it a test of *Oda's conjecture*: on a smooth projective toric variety, $H^0(\mathcal{O}(\mathbf{d})) \otimes H^0(\mathcal{O}(A)) \to H^0(\mathcal{O}(\mathbf{d} + A))$ is surjective for $\mathbf{d}$ nef and A ample [119]. It is known for surfaces [72] and for threefolds up to finitely many ample classes [138], but open in general; with $\mathbf{d} = kA$ it would make every smooth lattice polytope normal, itself open in dimension three [14].

Conjecture 4.5 (Cox–Mumford Multiplication)**.** *Let $\mathcal{F}$ be locally free on a smooth projective model X_i, and let $\mathbf{d} \in \mathcal{R}_i(\mathcal{F})$. For every ample Cartier divisor A on X_i, the multiplication map $H^0(X_i, \mathcal{F}(\mathbf{d})) \otimes H^0(X_i, \mathcal{O}_{X_i}(A)) \to H^0(X_i, \mathcal{F}(\mathbf{d} + A))$ is surjective.*

By Lemma 4.3, the case $\mathcal{F} = \mathcal{O}_{X_i}$ implies Oda's conjecture. A successful extension of Mumford's theorem to $D_{\mathrm{Cox}}(X)$ would prove it; a counterexample to Conjecture 4.5 would show where the heights fail to capture positivity. As evidence, I have proved Conjecture 4.5 for every split projective bundle over $\mathbb{P}^m$ using Cox-height vanishings in a mapping cone of base and fiber evaluation sequences (cf. [113]).

For general smooth toric models, the target is a finite resolution of the evaluation kernel $\mathcal{M}_A$ by bundles $\mathcal{O}_{X_i}(\beta)$ with $\nu(\beta) \leq p + 1$ in homological degree p, uniformly in A. Cox regularity would then force $H^1(X_i, \mathcal{F}(\mathbf{d}) \otimes \mathcal{M}_A) = 0$, proving Conjecture 4.5. The Cox diagonal applied to $\mathcal{M}_A$ gives such a resolution except for degree-zero terms $T_\alpha \boxtimes T_\beta$ with $\alpha \neq 0$ (e.g. the label $(1, -1)$ in Figure 3), which reintroduce multiplication maps and carry Oda's difficulty. When 0 is the only height-zero label, as on products of projective spaces, the argument should go through; I will prove this case first. Intermediate goals are Problem 4.2, Question 4.4 for Hirzebruch surfaces and scrolls, and a computer search for counterexamples on smooth toric threefolds.

5. Extending Computational Algebra Software

Computation drives recent progress in multigraded commutative algebra and toric geometry. Building on the PI's track record, each project yields a community tool supporting this proposal.

5.1 The `toricStacks` Package for *Macaulay2* A lack of software is a major bottleneck for studying homological invariants of stacks. A general framework is out of reach, but toric stacks correspond to *stacky fans*, whose well-developed combinatorics makes computation feasible [81, 82, 24]; yet no implementation exists. Given the success of *NormalToricVarieties* [127], a `toricStacks` package would benefit the community and directly support the experiments in §2.

Research Problem 5.1. *Develop a Macaulay2* [85] *package for computations with toric stacks.*

The package will (i) construct toric stacks from stacky fans and from families such as weighted projective stacks and root stacks, (ii) compute the Cox ring with its $\mathrm{Cl}(\mathcal{X})$-grading and irrelevant ideal, and (iii) support coherent sheaves, including sheaf cohomology via local cohomology [70]. This is joint work in progress with M. Banks and J. Cobb; (i) is complete and (ii) is underway.

5.2 A Database of Human-Generated Examples in Algebra Good examples drive conjectures, theorems, and algorithms. Thousands of expert-chosen rings, ideals, and modules lie scattered, largely untapped, across *Macaulay2*'s open-source documentation and tests.

Research Problem 5.2. *Build a Macaulay2 package collecting the algebraic objects in its source and documentation into a queryable database with provenance and invariants.*

It will parse and rerun installed packages' `EXAMPLE`/`TEST` blocks in a sandbox, recording directly callable objects with source and authorship. Much of the parser exists; execution and curation remain. This conjecture testbed and benchmark suite meets growing demand for mathematician-built AI/ML on mathematician-owned, fully attributed data [54], complementing curated repositories [131, 80].

6. Broader Impacts

6.1 Organizing. Building on §1.5, since 2022 I have organized *Gems of Combinatorics II* (30 participants), *AGNES* (150), and *BATMOBILE 2024 & 2025* (50/30). I co-organize Dartmouth's *Algebra & Number Theory* seminar. *Future Plans:* With co-organizers, I have proposed a *Thematic Semester in Combinatorial Algebraic Geometry* to the Fields Institute for 2030. I will host the next *Gems of Commutative Algebra* at Dartmouth in fall 2028 and write a *Notices of the AMS* article on organizing early-career conferences. J. Voight and I are in the planning stages of a Macaulay2+Magma workshop on the future of mathematical software.

6.2 Open Educational Resources. In fall 2026 I am lecturing for the Preliminary Arizona Winter School, a nine-week virtual course with 300 students (200 undergraduate and master's students); all materials (videos and 100 pages of inquiry-based notes and problems) are freely available. Building on the work of Karen Smith, I designed an inquiry-based commutative algebra course of 16 worksheets (200+ pages), all on my website. *Future Plans:* I will develop these materials into an open-access textbook introduction to algebraic geometry with Macaulay2 exercises.

6.3 Open Computational Resources. I created *Syzygy Data*, a public database of large syzygy computations, and co-authored four Macaulay2 packages. *Future Plans:* I will lead a group developing `toricStacks` at the 2027 *Macaulay2* conference at Auburn.

6.4 Campus Community. I am the faculty advisor to Dartmouth's *AWM Student Chapter* and work with the campus *oSTEM* chapter to broaden participation in STEM. In fall 2025 I founded *Unwritten Rules*, a graduate seminar that makes the hidden curriculum of an academic career explicit. *Future Plans:* I will extend *Unwritten Rules* to Dartmouth postdocs and use participant feedback to refine the seminar.

6.5 National & International Service. I have worked with the *AWM* Executive Committee to expand support for a broader range of mathematicians. From 2023 until its dissolution in spring 2025, I served on SLMath's *Committee on Women in Mathematics*. I now serve on the *AMS Committee on Engagement, Participation, & Advancement* and the *AMS Simons Travel Grant Committee*. In January 2026 I began a three-year term on the Board of Trustees of *Spectra*.

6.6 Mentoring. I advise three Ph.D. students on problems central to this proposal: Will D. (algebraic geometry + AI/ML), Jacob L-D. (regularity of torus-invariant subvarieties), and Alex M. (positivity of the BT collection). At MREG 2023 I began mentoring two graduate students on a project that led to [38]. In summer 2026 I mentored three undergraduates through *Count Me In* (organized by M. Hering), and since fall 2024 I have mentored two undergraduates. *Future Plans:* I will organize an undergraduate research program through Dartmouth's SHUR program in 2029.

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SUMMARY PROPOSAL BUDGET

YEAR 1

FOR NSF USE ONLY		
PROPOSAL NO.	DURATION (months)	
	Proposed	Granted
AWARD NO.		

ORGANIZATION: Dartmouth College

PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR: Juliette Bruce

A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)	NSF Funded Person-months CAL	ACAD	SUMR	Funds Requested By proposer	Funds granted by NSF (if different)
1. Juliette Bruce - Principal Inv	2.0			27,370	
2.					
3.					
4.					
5.					
6. ( ) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)	0.0			0	
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)	2.0			27,370	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)					
1. (0) POST DOCTORAL SCHOLARS	0.0			0	
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)	0.0			0	
3. (1) GRADUATE STUDENTS				51,940	
4. (1) UNDERGRADUATE STUDENTS				2,188	
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)				0	
6. (0) OTHER				0	
TOTAL SALARIES AND WAGES (A + B)				81,498	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)				10,324	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)				91,822	
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING $10,000.)					
TOTAL EQUIPMENT				0	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)				6,176	
2. INTERNATIONAL				5,000	
F. PARTICIPANT SUPPORT COSTS					
1. STIPENDS $ 0					
2. TRAVEL 0					
3. SUBSISTENCE 0					
4. OTHER 0					
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS				0	
G. OTHER DIRECT COSTS					
1. MATERIALS AND SUPPLIES				0	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION				0	
3. CONSULTANT SERVICES				0	
4. COMPUTER SERVICES				5,000	
5. SUBAWARDS				0	
6. OTHER				6,025	
TOTAL OTHER DIRECT COSTS				11,025	
H. TOTAL DIRECT COSTS (A THROUGH G)				114,023	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 63.92, Base:114023)					
TOTAL INDIRECT COSTS (F&A)				72,884	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)				186,907	
K. FEE				0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)				186,907	

M. COST SHARING PROPOSED LEVEL $ 0 AGREED LEVEL IF DIFFERENT $

PI/PD NAME: Juliette Bruce

ORG. REP. NAME*: Arlanda Erzen Lumbra

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INDIRECT COST RATE VERIFICATION		
Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 2

FOR NSF USE ONLY		
PROPOSAL NO.	DURATION (months)	
	Proposed	Granted
AWARD NO.		

ORGANIZATION: Dartmouth College

PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR: Juliette Bruce

A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)	NSF Funded Person-months CAL	ACAD	SUMR	Funds Requested By proposer	Funds granted by NSF (if different)
1. Juliette Bruce - Principal Inv	2.0			28,739	
2.					
3.					
4.					
5.					
6. ( ) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)	0.0			0	
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)	2.0			28,739	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)					
1. (0) POST DOCTORAL SCHOLARS	0.0			0	
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)	0.0			0	
3. (1) GRADUATE STUDENTS				54,537	
4. (1) UNDERGRADUATE STUDENTS				2,297	
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)				0	
6. (0) OTHER				0	
TOTAL SALARIES AND WAGES (A + B)				85,573	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)				10,840	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)				96,413	
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING $10,000.)					
TOTAL EQUIPMENT				0	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)				6,176	
2. INTERNATIONAL				5,000	
F. PARTICIPANT SUPPORT COSTS					
1. STIPENDS $ 0					
2. TRAVEL 0					
3. SUBSISTENCE 0					
4. OTHER 0					
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS				0	
G. OTHER DIRECT COSTS					
1. MATERIALS AND SUPPLIES				0	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION				0	
3. CONSULTANT SERVICES				0	
4. COMPUTER SERVICES				0	
5. SUBAWARDS				0	
6. OTHER				6,325	
TOTAL OTHER DIRECT COSTS				6,325	
H. TOTAL DIRECT COSTS (A THROUGH G)				113,914	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 64.0, Base:113914)					
TOTAL INDIRECT COSTS (F&A)				72,905	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)				186,819	
K. FEE				0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)				186,819	

M. COST SHARING PROPOSED LEVEL $ 0 AGREED LEVEL IF DIFFERENT $

PI/PD NAME: Juliette Bruce

ORG. REP. NAME*: Arlanda Erzen Lumbra

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INDIRECT COST RATE VERIFICATION		
Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

Cumulative

FOR NSF USE ONLY		
PROPOSAL NO.	DURATION (months)	
	Proposed	Granted
AWARD NO.		

ORGANIZATION: Dartmouth College

PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR: Juliette Bruce

A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)	NSF Funded Person-months CAL	ACAD	SUMR	Funds Requested By proposer	Funds granted by NSF (if different)
1. Juliette Bruce - Principal Inv	4.0			56,109	
2.					
3.					
4.					
5.					
6. ( ) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)					
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)	4.0			56,109	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)					
1. (0) POST DOCTORAL SCHOLARS	0.0			0	
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)	0.0			0	
3. (2) GRADUATE STUDENTS				106,477	
4. (2) UNDERGRADUATE STUDENTS				4,485	
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)				0	
6. (0) OTHER				0	
TOTAL SALARIES AND WAGES (A + B)				167,071	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)				21,164	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)				188,235	
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING $10,000.)					
TOTAL EQUIPMENT				0	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)				12,352	
2. INTERNATIONAL				10,000	
F. PARTICIPANT SUPPORT COSTS					
1. STIPENDS $ 0					
2. TRAVEL 0					
3. SUBSISTENCE 0					
4. OTHER 0					
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS				0	
G. OTHER DIRECT COSTS					
1. MATERIALS AND SUPPLIES				0	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION				0	
3. CONSULTANT SERVICES				0	
4. COMPUTER SERVICES				5,000	
5. SUBAWARDS				0	
6. OTHER				12,350	
TOTAL OTHER DIRECT COSTS				17,350	
H. TOTAL DIRECT COSTS (A THROUGH G)				227,937	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)					
TOTAL INDIRECT COSTS (F&A)				145,789	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)				373,726	
K. FEE				0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)				373,726	

M. COST SHARING PROPOSED LEVEL $ 0 AGREED LEVEL IF DIFFERENT $

PI/PD NAME: Juliette Bruce

ORG. REP. NAME*: Arlanda Erzen Lumbra

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INDIRECT COST RATE VERIFICATION		
Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

BUDGET JUSTIFICATION

Dartmouth College defines a year as 12 calendar months. Salaries for all personnel are increased by 5% annually effective July 1st at the beginning of Dartmouth's fiscal year. The personnel funds requested are based on the individual annual salary multiplied by proposed effort.

A. SENIOR PERSONNEL

Juliette Bruce, Principal Investigator, (2.00 summer month per year) will oversee all aspects of the project and perform research as described in the project narrative. This proposed level of commitment is appropriate to the scope of work and in order to fulfill the research objectives.

B. OTHER PERSONNEL

TBD, Graduate Student, (12.00 calendar months per year) will perform research in topics related to the projects included in this proposal. In particular, developing computer software that helps compute various combinatorial invariants related to this project.

TBD, Undergraduate Student, will work on data generation, collection, and curation and website maintenance for a database connected to research in topics related to the projects included in this proposal. We estimate the undergraduate student will spend a total of ~100 hours per year working on the project.

C. FRINGE BENEFITS

The fringe benefit rate on salary support for all personnel (excluding the Graduate Student and Undergraduate Student) is budgeted at 37.0%. The fringe rate on the Undergraduate Student costs is 9% for all project years. There are no fringe benefit costs for the Graduate Student.

E. TRAVEL

1. Domestic: Travel funds of $4,000 per year are requested to allow the research team to present results, build professional networks, and obtain valuable exposure to the research community at a professional conference such as Western Algebraic Geometry Seminar, the Joint Mathematics Meetings, and Sectionals of the American Mathematical Society. Travel will be for 1 person each year to attend a 6 days/5 nights conference. Costs follow standard college travel policies and are estimated as follows:

Cost	Details	Qty	Rate	Per Person	Per Year
Airfare	Flight to/from BOS	1	$700	$700	$700
Conference Registration	Per Registration	1	$1,000	$1,000	$1,000
Lodging	Per Night	5	$320	$1,600	$1,600
Per Diem Meals/Incidentals	Per Day as per GSA	6	$92	$552	$552
Ground transportation	Bus, parking, taxi, etc	1	$148	$148	$148
		Total Travel		$4,000	$4,000

Travel funds of $2,176 per year are also requested to allow the research team to meet with collaborators. Possible meeting locations include U. of Minnesota, U. of San Diego, UC Berkeley, U. of Wisconsin, Texas A&M, Washington U., St. Louis, Cornell, and San Francisco State. Travel will be for 1 person each year for 4 days/3 nights conference. Costs follow standard college travel policies and are estimated as follows:

Cost	Details	Qty	Rate	Per Person	Per Year
Airfare	Flight to/from BOS	1	$700	$700	$700
Lodging	Per Night	3	$320	$960	$960
Per Diem Meals/Incidentals	Per Day as per GSA	4	$92	$368	$368
Ground transportation	Bus, parking, taxi, etc	1	$148	$148	$148
		Total Travel		$2,176	$2,176

2. Foreign: Travel funds of $5,000 per year are requested to allow the research team to present results, build professional networks, and obtain valuable exposure to the research community at a professional international conference such as Oberwolfach, MEGA, and the SIAM Applied Algebra Conference. Travel will be for 1 person each year for 6 days/5 nights. Costs follow standard college travel policies and are estimated as follows:

Cost	Details	Qty	Rate	Per Person	Per Year
Airfare	Flight to/from BOS	1	$900	$900	$900
Conference Registration	Per Registration	1	$975	$975	$975
Lodging	Per Night	5	$375	$1,875	$1,875
Per Diem Meals/Incidentals	Per Day as per GSA	6	$184	$1,104	$1,104
Ground transportation	Bus, parking, taxi, etc	1	$146	$146	$146
		Total Travel		$5,000	$5,000

G. OTHER DIRECT COSTS

4. Computer Services: Funds of $5,000 are requested in year 1 for UM hosting of high-speed multi-parallel computations run on a computer maintained by the PI. Costs are estimated based on the current market prices of high-performance desktop computers/servers and the PI's experience.

5. OTHER:
   a. Health Insurance & Health Access Fees: Funds are requested in support of the Graduate Student at $5,433 for health insurance and $592 for fees in year 1 and increases by 5% annually.

I. INDIRECT COSTS

Project Year	Yr1	Yr2
MTDC Base	$114,023	$113,914
Total Indirect Costs Requested	$72,884	$72,905

In accordance with Dartmouth's negotiated F&A Rate Agreement with the Department of Health and Human Services (DHHS) dated 04/09/2026, the predetermined on-campus research rate is 63.5% from 7/1/2025 to 6/30/2027 and 64% from 7/1/2027 onward, calculated on a modified total direct cost (MTDC) base (MTDC = total direct costs excluding equipment, capital expenditures, charges for patient care, rental costs, tuition remission, scholarships and fellowships, participant support costs and the portion of each subaward in excess of $25,000). The cognizant federal agency is DHHS, Olulola Oluborode, Director, Division of Cost Allocation Services, 301-492-4624

Facilities, Equipment, and Other Resources

Laboratory:
N/A

Clinical:
N/A

Animal:
N/A

Computer:
Dartmouth College and the Department of Mathematics at Dartmouth College have made significant investments in having the infrastructure necesary for large scale computation. Dartmouth Research computing has ,"A total of 6000+ CPU cores, 120,000 GPU cores, 12+TB of memory, and 1 PB of storage across all systems" including many high-performance computing nodes. The Department of Mathematics has a number of its own servers dedicated to computational research. Dartmouth College also provides software licenses and access to Magma, Mathematica, Maple, R, Python, and other applications.

Space:
All faculty, graduate students, and visitors are provided with sufficient office space to conduct their teaching and research.

IDENTIFYING INFORMATION:

NAME: Bruce, Juliette

ORCID iD: https://orcid.org/0000-0001-9766-7896

POSITION TITLE: Assistant Professor

PRIMARY ORGANIZATION AND LOCATION: Dartmouth College, Hanover, NH, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Brown University, Providence, Rhode Island, United States	Postdoctoral Fellow	08/2022 - 06/2024	Mathematics
University of California, Berkeley, Berkeley, California, United States	Postdoctoral Fellow	08/2020 - 07/2022	Mathematics
Mathematical Sciences Research Institute, Berkeley, California, United States	Postdoctoral Fellow	08/2020 - 05/2021	Mathematicas
University of Wisconsin–Madison, Madison, Wisconsin, US	Ph.D	09/2014 - 08/2020	Department of Mathematics
University of Wisconsin–Madison, Madison, Wisconsin, US	Masters of Arts	09/2014 - 05/2016	Department of Mathematics
University of Michigan–Ann Arbor, Ann Arbor, Michigan, US	Bachelors of Science	09/2010 - 04/2014	Department of Mathematics / Department of Political Science

Appointments and Positions

2024 - present Assistant Professor, Dartmouth College, Hanover, NH, United States
2025 - 2025 Long-Term Visiting Member, Fields Institute, Toronto, Ontario, Canada
2024 - present Assistant Professor, Dartmouth College, Mathematics, Hanover, New Hampshire, United States
2014 - 2020 Teaching Assistant / Research Assistant, University of Wisconsin - Madison, Department of Mathematics, Madison, WI, US

Products

Products Most Closely Related to the Proposed Project

1. Madeline Brandt, Juliette Bruce, Melody Chan, Margarida Melo, Gwyneth Moreland, Corey Wolfe. On the top-weight rational cohomology of $\mathcal{A}_g$. Geometry & Topology.

2024 March. DOI: 10.48550/ARXIV.2012.02892

2. Bruce J, Erman D. A probabilistic approach to systems of parameters and Noether normalization. Algebra & Number Theory. 2019 December 7; 13(9):2081-2102. Available from: https://msp.org/ant/2019/13-9/p05.xhtml DOI: 10.2140/ant.2019.13.2081
3. Bruce J. Asymptotic syzygies for products of projective spaces. Journal of Algebra. 2024 July; 649:347-391. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0021869324001479 DOI: 10.1016/j.jalgebra.2024.03.013
4. Bruce J, Cranton Heller L, Sayrafi M. Characterizing multigraded regularity and virtual resolutions on products of projective spaces. Advances in Mathematics. 2026 August; 500:111049-. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0001870826002719 DOI: 10.1016/j.aim.2026.111049
5. Juliette Bruce, Lauren Cranton Heller, Mahrud Sayrafi. Bounds on Multigraded Regularity. 2022. Available from: https://arxiv.org/abs/2208.11115 DOI: 10.48550/ARXIV.2208.11115

Other Significant Products, Whether or Not Related to the Proposed Project

1. Bruce J, Erman D, Goldstein S, Yang J. Conjectures and Computations about Veronese Syzygies. Experimental Mathematics. 2018 June 21; 29(4):398-413. Available from: https://www.tandfonline.com/doi/full/10.1080/10586458.2018.1474506 DOI: 10.1080/10586458.2018.1474506
2. Juliette Bruce. The quantitative behavior of asymptotic syzygies for Hirzebruch surfaces. Journal of Commutative Algebra. 2022 March. DOI: 10.1216/jca.2022.14.19
3. Eran Assaf, Madeline Brandt, Juliette Bruce, Melody Chan, Raluca Vlad. Tropicalizations of locally symmetric varieties. 2025. Available from: https://arxiv.org/abs/2505.10504 uri: https://arxiv.org/abs/2505.10504
4. Juliette Bruce. Eventual Nonstandard Koszulness Fails for Veronese Subrings of Weighted Polynomial Rings. 2026. Available from: https://arxiv.org/abs/2608.23913 DOI: 10.48550/ARXIV.2608.23913
5. Juliette Bruce, Lauren Cranton Heller, Mahrud Sayrafi, Alexandra Seceleanu. Seshadri Regions and the Asymptotic Shape of Multigraded Regularity. 2025. Available from: https://arxiv.org/abs/2512.05289 DOI: 10.48550/ARXIV.2512.05289

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Bruce, Juliette in SciENcv on 2026-09-10 04:05:46

Other Personnel Biographical Information

Data Not Available

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Bruce, Juliette

PERSISTENT IDENTIFIER (PID) OF THE SENIOR/KEY PERSON: https://orcid.org/0000-0001-9766-7896

*POSITION TITLE: Assistant Professor

*ORGANIZATION AND LOCATION: Dartmouth College, Hanover, NH, United States

Proposals/Active Projects

***Proposal/Active Project Title:** Multigraded Homological Algebra & Geometry

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NSF

***Primary Place of Performance:** Dartmouth College

***Proposal/Active Project Start Date: (MM/YYYY):** 05/2027

***Proposal/Active Project End Date: (MM/YYYY):** 04/2029

***Total Anticipated Proposal/Project Amount:** $373,726

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2028	2
2029	2

***Overall Objectives:**

This proposal advances the study of multigraded homological algebra, algebraic geometry, and combinatorics.

***Statement of Potential Overlap:**

This proposal is the current proposal under review.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Bruce, Juliette in SciENcv on 2026-10-07 09:27:07

Table 1

1	Your Name:	Your Organizational Affiliation(s), last 12 mo	Last Active Date
	Bruce, Juliette E	Dartmouth College	

Table 2

2	Name:	Type of Relationship	Optional (email, Department)	Last Active Date
R	Newton, Kit	Family	knewton@dvc.edu	

Table 3

3	Advisor/Advisee Name:	Organizational Affiliation	Optional (email, Department)
G	Erman, Daniel	University of Hawaii	erman@math.hawaii.edu
T	DeGroot, William	Dartmouth College	william.h.degroot.gr@dartmouth.edu

Table 4

4	Name:	Organizational Affiliation	Optional (email, Department)	Last Active Date
C	Ardila, Federico	San Francisco State University	federico@sfsu.edu	06/01/24
A	Assaf, Eran	Massachusetts Institute of Technology	eranasaf@mit.edu	
C	Banks, Maya	University of Illinois, Chicago	mayadb@uic.edu	
C	Beck, Matthias	San Francisco State University	mattbeck@sfsu.edu	06/01/24
C	Berkesch, Christine	University of Minnesota - Twin Cities	cberkesc@umn.edu	
A	Bonato, Anthony	Ryerson University	abonato@ryerson.ca	01/01/24
C	Boocher, Adam	University of San Diego	aboocher@sandiego.edu	06/01/24
A	Brandt, Madeline	Vanderbilt Universtiy	madeline.v.brandt@vanderbilt.edu	
C	Bucciarelli, Jacob	Kansas State University	jbucciarelli@ksu.edu	
C	Buckmire, Ron	Ocidental College	ron@oxy.edu	01/01/24
A	Cavalieri, Renzo	Colorado State University	renzo@math.colostate.edu	
A	Chan, Melody	Brown University	melody_chan@brown.edu	
A	Cobb, John	Auburn University	jdcobb3@gmail.com	
A	Corey, Daniel	University of Nevada - Las Vegas	daniel.corey@unlv.edu	
A	Cranton Heller, Lauren	University of Nebraska - Lincoln	lheller2@unl.edu	
A	Eisenbud, David	University of California - Berkeley	de@math.berkeley.edu	07/31/22
C	Erman, Daniel	University of Hawaii	erman@math.hawaii.edu	
C	Hering, Milena	Edinburgh University	m.hering@ed.ac.uk	
C	Hosten, Serkan	San Francisco State University	serkan@sfsu.edu	06/01/24
C	Huang, Daoji	Institute for Advanced Study	dhuang@ias.edu	

C	Kelly, Tyler, L	University of Birmingham	t.kelly.1@bham.ac.uk	08/01/22
C	Klein, Patricia	Texas A&M University	pjklein@tamu.edu	06/01/25
A	Maclagan, Diane	University of Warwick	D.Maclagan@warwick.ac.uk	
A	Manon, Christopher	University of Kentucky	christopher.manon@uky.edu	
C	Maclagan, Diane	University of Warwick	D.Maclagan@warwick.ac.uk	
A	Melo, Margarida	Roma Tre University	melo@mat.uniroma3.it	07/01/22
A	Moreland, Gwyneth	Harvard University	gwynm@math.harvard.edu	07/01/22
C	Ramkumar, Ritvik	University of Notre Dame	rramkuma@nd.edu	07/31/22
C	Ross, Dustin	San Francisco State University	rossd@sfsu.edu	
A	Sayrafi, Mahrud	McMaster University	mahrud@mcmaster.ca	
C	Seceleanu, Alexandra	University of Nebraska - Lincoln	aseceleanu@unl.edu	
A	Sayrafi, Mahrud	University of Tulane	cwolfe@tulane.edu	
A	Vlad, Raluca	Brown University	raluca_vlad@brown.edu	
C	Voight, John	Dartmouth College	jvoight@gmail.com	08/01/22

Table 5

5	Name:	Organizational Affiliation	Journal/Collection	Last Active Date
B	Huybrechts, Daniel	Universität Bonn	Journal für die reine und angewandte Mathematik	01/01/26
B	Payne, Sam	University of Michigan	Selecta Mathematica	07/01/25
B	Berkesch, Christine	University of Minnesota	La Matematica	

Synergistic Activities

1. **Open software for multigraded algebra.** The PI co-developed *VirtualResolutions*, an open-source package distributed with Macaulay2 for constructing and checking virtual resolutions and computing multigraded regularity on products of projective spaces. Its documented algorithms and worked examples give researchers and students practical access to methods connecting commutative algebra and algebraic geometry. Accompanied by a peer-reviewed article in the *Journal of Software for Algebra and Geometry* (2020), the package translates theoretical constructions into reusable tools for investigating examples and testing conjectures.

2. **Advanced mathematical training through PAWS.** The PI serves as lecturer for *Algebraic Geometry via Computation* in the 2026 Preliminary Arizona Winter School (PAWS), a nine-week virtual program with over 300 participants, including approximately 200 students who have not yet entered Ph.D. programs. She prepared 100 pages of lecture notes on polynomial ideals,

3. **Research mentoring through MREG.** The PI led a group of four early-stage graduate students at the 2023 Midwest Research Experience for Graduates at the University of Michigan, introducing them to research on matroids at the interface of algebraic geometry and combinatorics. The collaboration that began there led to the 2026 preprint *Explorations of Matroid Complexes*, coauthored with Jacob Bucciarelli and Bailee Zacovic, and publicly available computational code and data. The project linked introductory research training with sustained collaboration and the creation of reusable computational resources.

4. **Plenary lecture at the Fields Institute.** The PI delivered the plenary lecture *Syzygies Beyond Projective Space* at the Fields Institute's Spec($\overline{\mathbb{Q}}(2\pi i)$) conference in June 2024. Addressing a general algebraic geometry audience, she explained how homological methods over multigraded polynomial rings extend the study of syzygies beyond projective space to toric varieties. The publicly available recording makes these methods accessible to researchers and students beyond the conference, supporting the exchange of ideas between commutative algebra and algebraic geometry.

5. **National professional leadership through Spectra.** As inaugural president of *Spectra*, the PI helped develop a national professional community supporting mathematicians and students across institutions and research areas. She helped establish the annual Spectra Lavender Lecture, co-organizing its inaugural presentation at the 2022 Joint Mathematics Meetings. The lecture recognizes contributions to mathematical research, education, and the profession, giving mathematicians a recurring platform to share their work and making these contributions visible to students and the wider mathematical community.

Data Management and Sharing Plan

Submitted to:
Directorate for Mathematical and Physical Sciences (MPS), MPS - Core Programs (MCP), ALGEBRA,NUMBER THEORY,AND COM

1. Mathematical Articles and Pre-prints

Data or Research Product Category General Information

Data or Research Product Category: **Mathematical Articles and Pre-prints**

Description: **All mathematical articles will be made publicly available via the arXiv pre-print server, before eventually being submitted to peer-reviewed journals.**

Access Policies and Limitations

Not applicable

Data Standards and Metadata

Data/Metadata Standard: **arXiv DOI**

Data or Research Product Provenance

Data Source: **Mathematics Research**

Public Archiving

Data Sharing Location: **arXiv**

Additional Details: **Pre-prints will be shared via the arXiv pre-print server, which is publicly available.**

Timeline for Public Accessibility

Data Sharing Timeline: **At time of publication**

Data Availability

Data Availability Period: **At least 3 years**

Will you adhere to the standard retention policy of the chosen repository: **Yes**

Accountability

Responsible Personnel: **Juliette Bruce**

Role: **Principal Investigator**

Organization Associated within the Proposal: **Dartmouth College**

Email Address: **juliette.bruce@dartmouth.edu**

2. Mathematical Software & Major Mathematical Computations

Data or Research Product Category General Information

Data or Research Product Category: **Mathematical Software & Major Mathematical Computations**

Description: **Any computer code written or examples generated, which might be interesting to others, will be made publicly accessible from the PI's website or GitHub repository.**

Access Policies and Limitations

Not applicable

Data Standards and Metadata

Data/Metadata Standard: **GitHub Link**

Data or Research Product Provenance

Data Source: **Mathematical Research**

Public Archiving

Data Sharing Location: **GitHub**

Additional Details: **Any computer code written or examples generated, which might be interesting to others, will be made publicly accessible from the PI's website or GitHub repository.**

Timeline for Public Accessibility

Data Sharing Timeline: **At time of publication**

Data Availability

Data Availability Period: **At least 3 years**

Will you adhere to the standard retention policy of the chosen repository: **Yes**

Accountability

Responsible Personnel: **Juliette Bruce**

Role: **Principal Investigator**

Organization Associated within the Proposal: **Dartmouth College**

Email Address: **juliette.bruce@dartmouth.edu**

MENTORING PLAN

Orientation: The PI will meet with all students and post-docs (none planned) associated with this grant to have a careful conversation surrounding the goals of this proposal, their career and educational goals, and the expectations for participating in these projections. These expectations and commitments from both the student/post-doc and the PI will be recorded in writing and frequently used as a benchmark to judge progress and make adjustments.

Professional Advising: In addition to the research advising the PI will provide students/post-docs with mentorship and advice on how they can move towards achieving their professional goals. This will include both informal advising and formal advising at scheduled monthly meetings specifically designated towards conversations around professional goals.

Preparing Publications: The PI will provide the students/postdocs with advice and suggestions for how to approach academic publishing, both in theory and in practice with any publications that may result from this work.

Networking: The PI will help students and post-docs identify and apply to attend conferences and workshops appropriate for building their professional and research networks.

Other Supplementary Documents

Data Not Available

List of Suggested Reviewers

Data Not Available

List of Reviewers Not to Include

Data Not Available